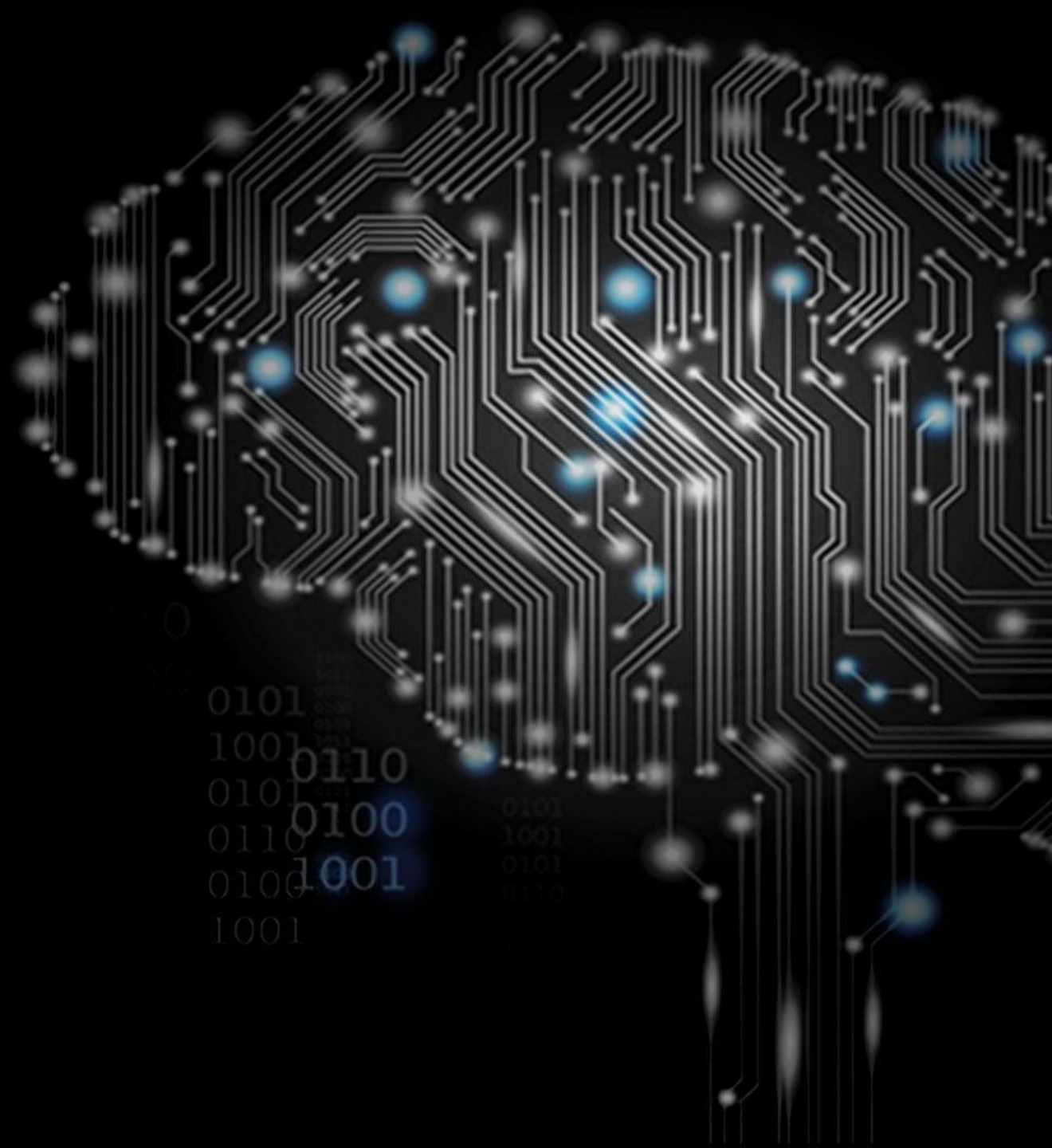


Introduction to Generative AI

1. The AI Landscape

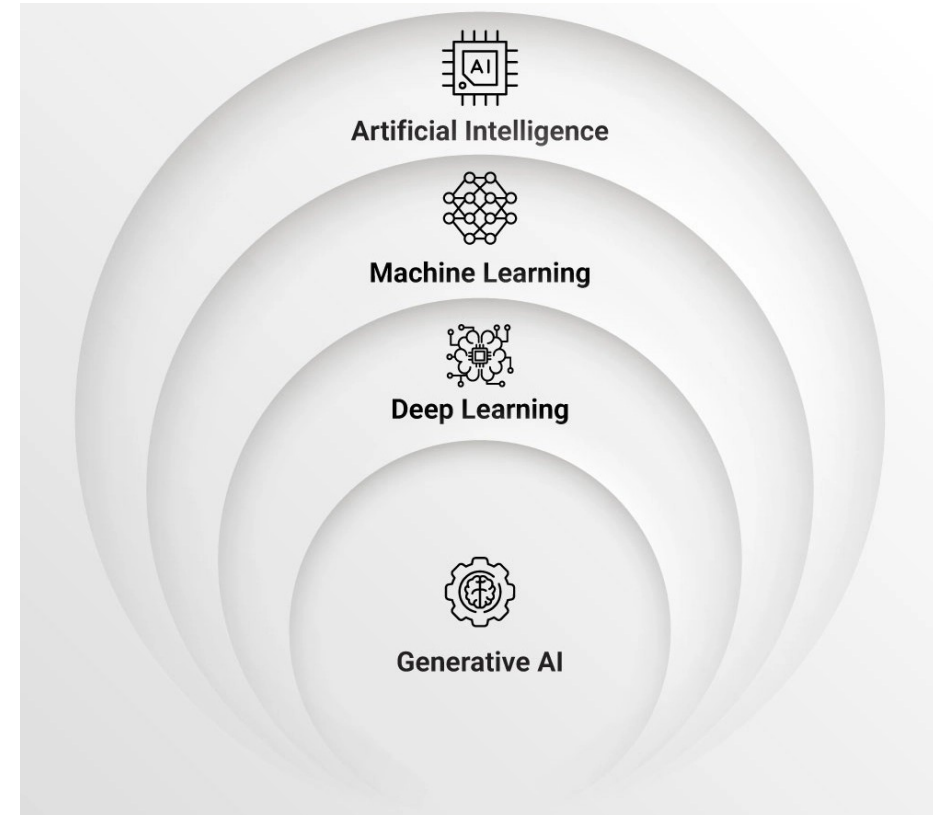


Module Topics

- Recent history of AI
- Classical machine learning
- Neural networks and deep learning
- Transformers and large language models
- Foundation models
- Major players in the industry
- Trends to watch for

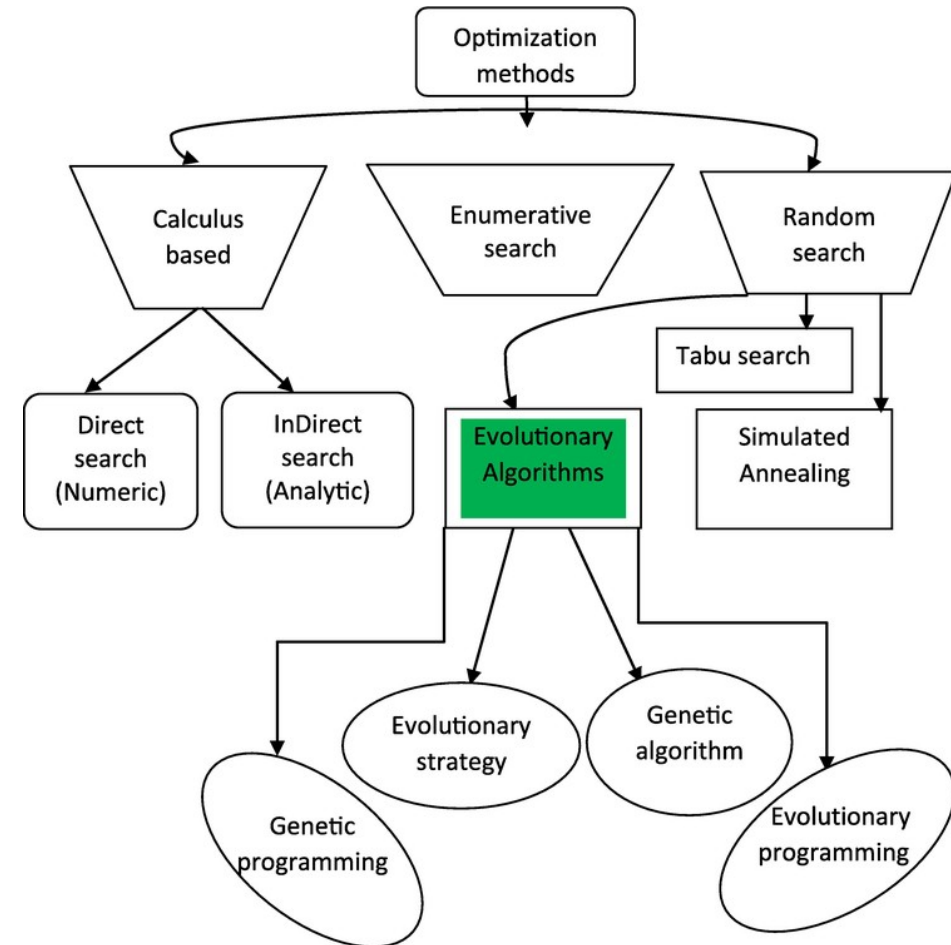
Defining AI

- AI includes a number of different fields of study
 - The current focus of AI is on machine learning, which includes LLMs and GAI
- There are other areas of AI that are not machine learning, for example
 - Cognitive computing
 - Artificial life and swarm intelligence
 - Evolutionary computing
 - Expert systems



Genetic and Evolutionary Computation

- Uses algorithms modelled after natural selection and evolution
 - Often used to solve problems that are too computationally expensive to solve otherwise
 - For example, PostgreSQL uses genetic computation to help optimize queries
- These are part of AI that focuses on developing optimal constraint based problems



Evolutionary Example

- The image shown is the 2006 NASA ST5 spacecraft antenna
 - Developed by an evolutionary algorithm
 - Its unusual shape is characteristic of products designed via evolutionary algorithms
 - Human antenna designers would never consider this sort of design
 - This design works very well in meeting the mission requirements constraints



Evolutionary Example

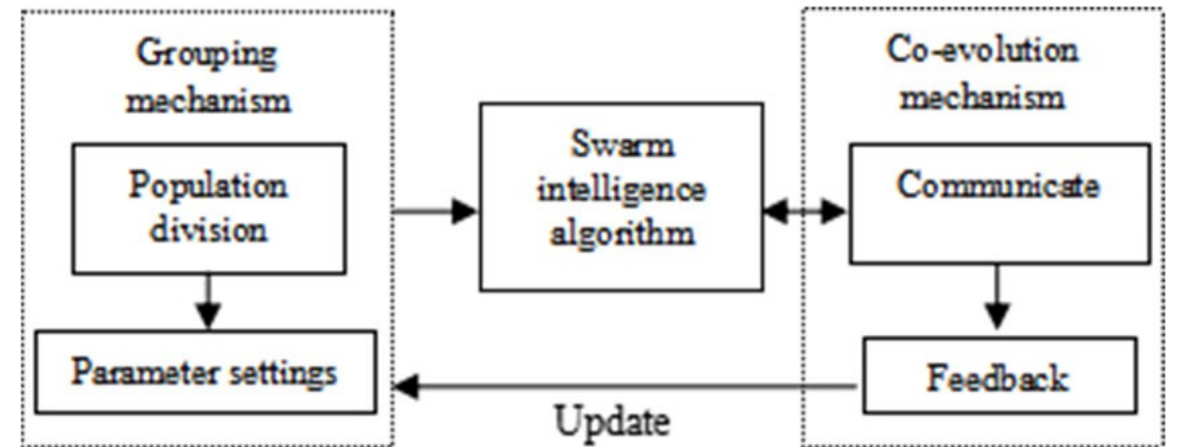
- The image shows a lightweight, lattice-like structural component for an extra-planetary lander
 - The structure has an organic, webbed geometry that differs significantly from traditional rigid, boxy aerospace frames
 - This is characteristic of generative design
 - It appears organic because evolutionary algorithms remove unnecessary material while reinforcing stress-bearing regions
 - Similar to how bones or natural forms evolve under environmental pressures.



NASA's Jet Propulsion Laboratory used generative design to develop a concept extraplanetary lander with lower mass and improved performance.

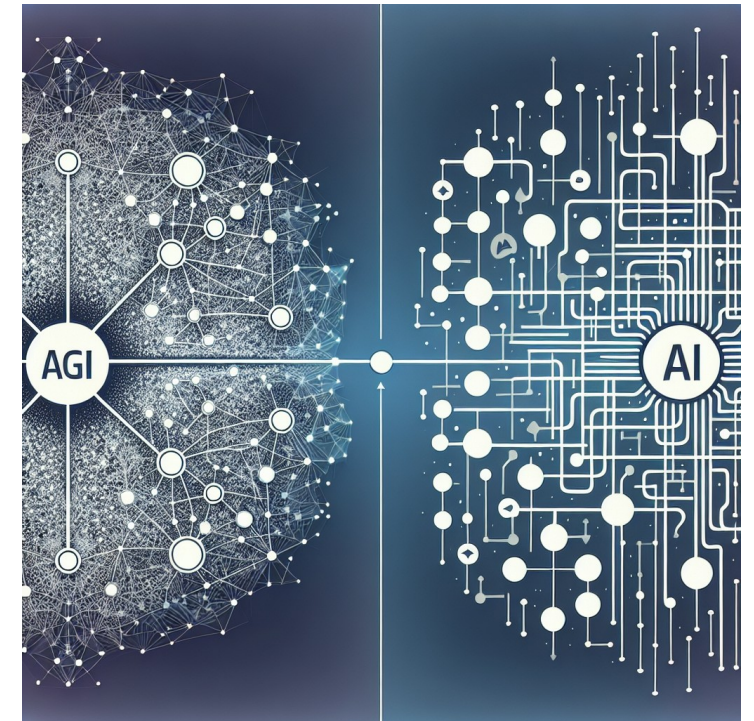
Swarm Intelligence Example

- Inspired by collective behavior in nature (ants, bees, birds, fish schools)
 - Intelligence emerges from many simple agents following local rules
 - Decentralized and self-organizing
 - Global behavior arises from local interactions
 - Highly adaptive, scalable, and robust while being tolerant to failure of individual agents
- Works well in complex, dynamic environments
 - Naturally suited for parallel computing
 - Scales efficiently with problem size
 - Useful when traditional optimization struggles
 - Aligns with decentralized AI and multi-agent systems



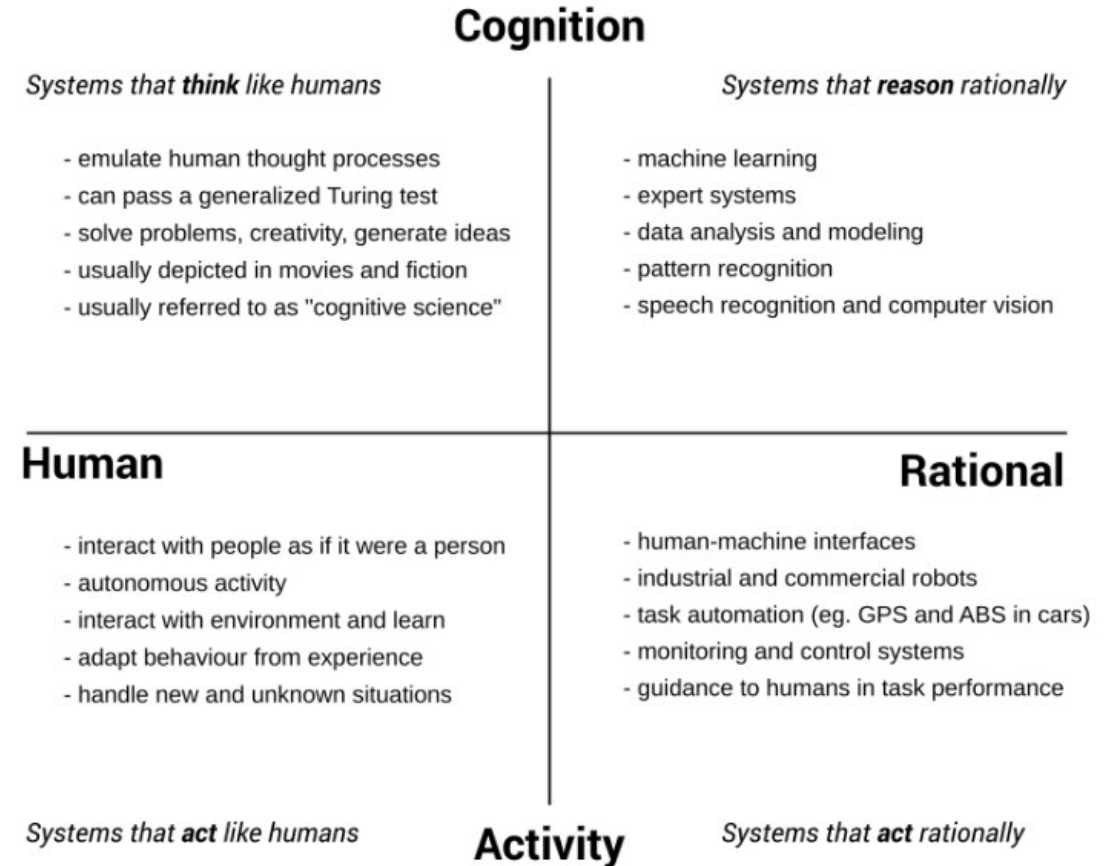
AGI and ANI

- Artificial general intelligence (AGI)
 - AI that possesses human-like intelligence and is capable of understanding and learning any intellectual task that a human being can
 - Often referred to as “strong AI” or “full AI”
 - Able to perform a wide range of tasks and adapt to new situations without being explicitly programmed to do so
 - Beyond any current capability we have
- Artificial narrow intelligence (ANI)
 - AI systems that are designed for specific tasks or functions.
 - Specialized and limited in scope
 - Focusing on solving particular problems or performing specific functions within a narrow domain

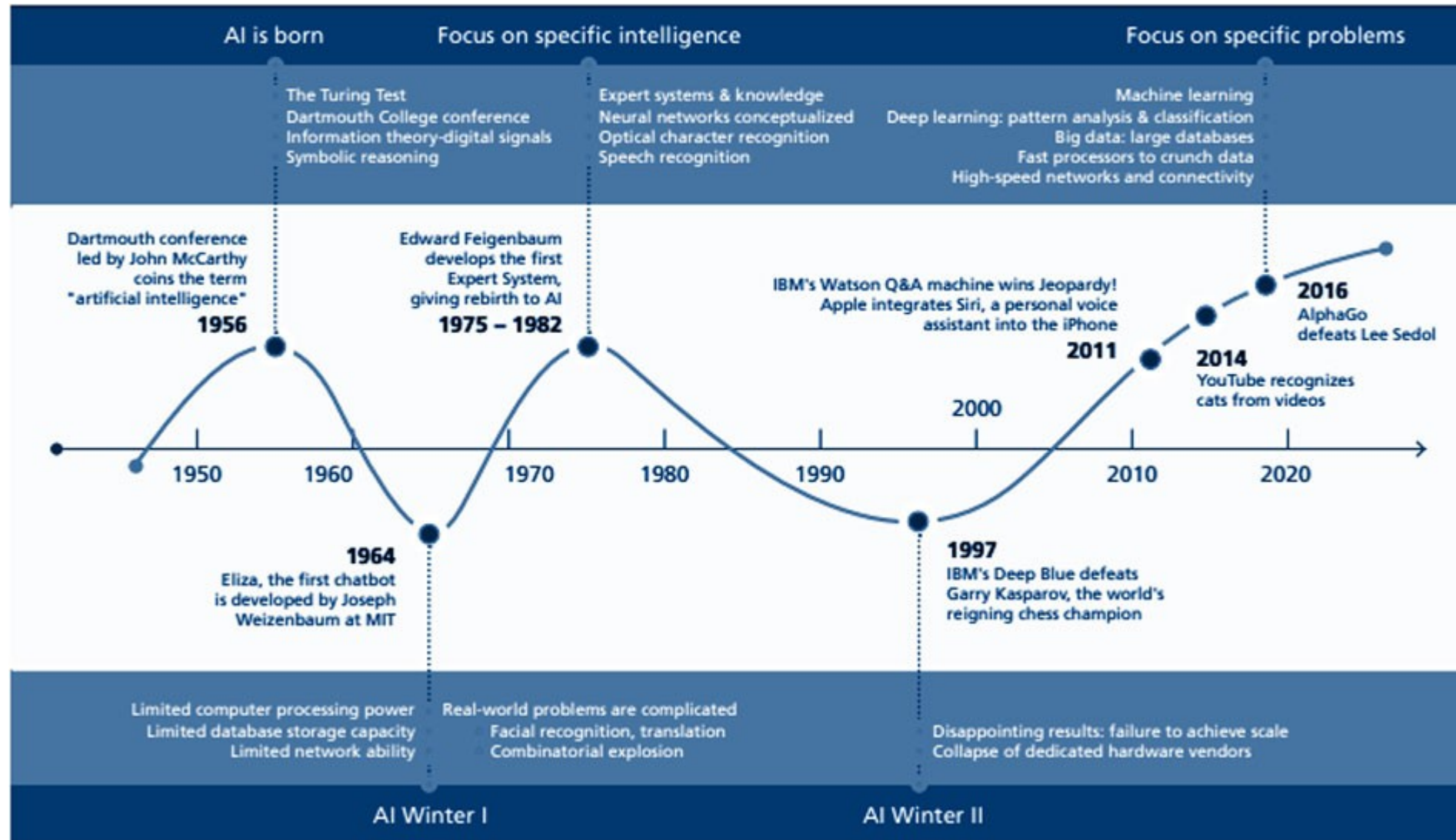


The Norvig Matrix

- Provides a general classification of the different types of AI based on their goals
 - The “Human” column represents the goals of AGI, which are currently unattainable
 - We do not understand human cognition and behaviour enough to simulate it
 - Posited that we have to “grow” AGI by raising it like a child
 - The “Rational” column represents what we currently do in AI
 - Systems that reason rationally produce results that are acceptable to humans, although the techniques used are not the same as people use
 - Systems that act rationally perform tasks in ways that are acceptable to humans but do not use the same processes as humans

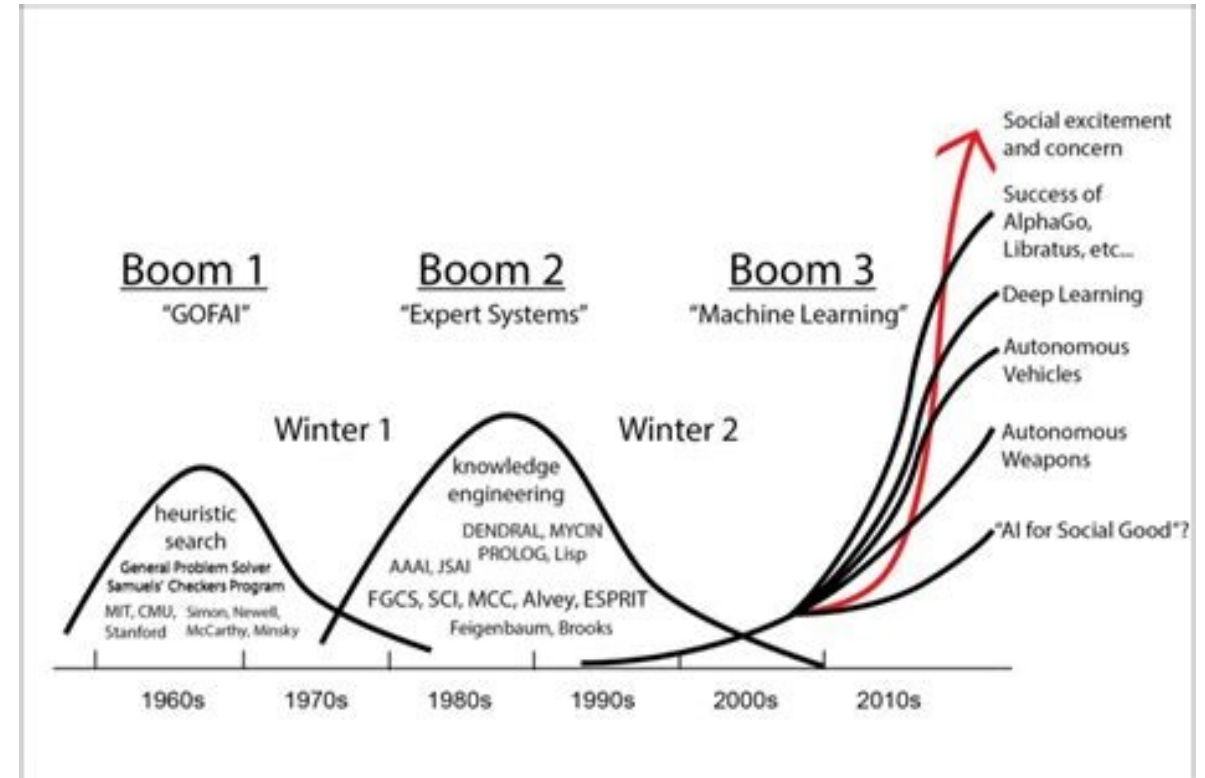


AI Timeline



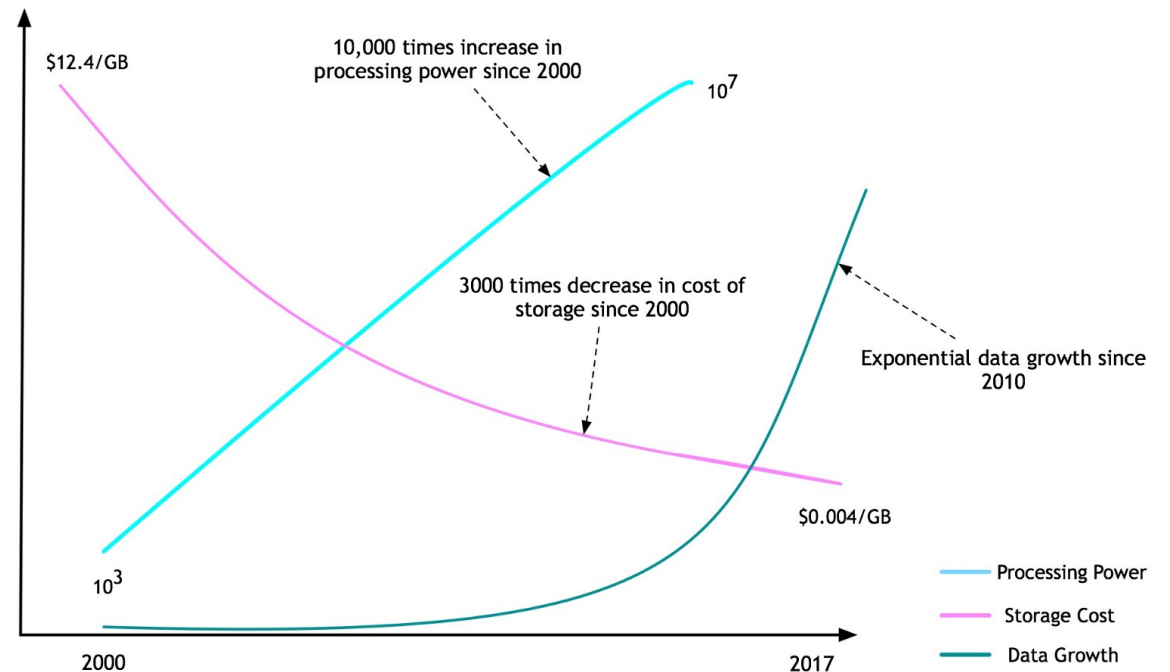
AI Cycles

- AI has been around since the 1940s
- There have been three main AI booms
 - Enabled by advances in hardware and availability of compute resources
 - Supported by investment and funding
- Each was followed by an AI winter
 - AI companies were not able to deliver on their promises
 - Funding dried up
 - AI research continued during these winters but there was little commercial or industrial development

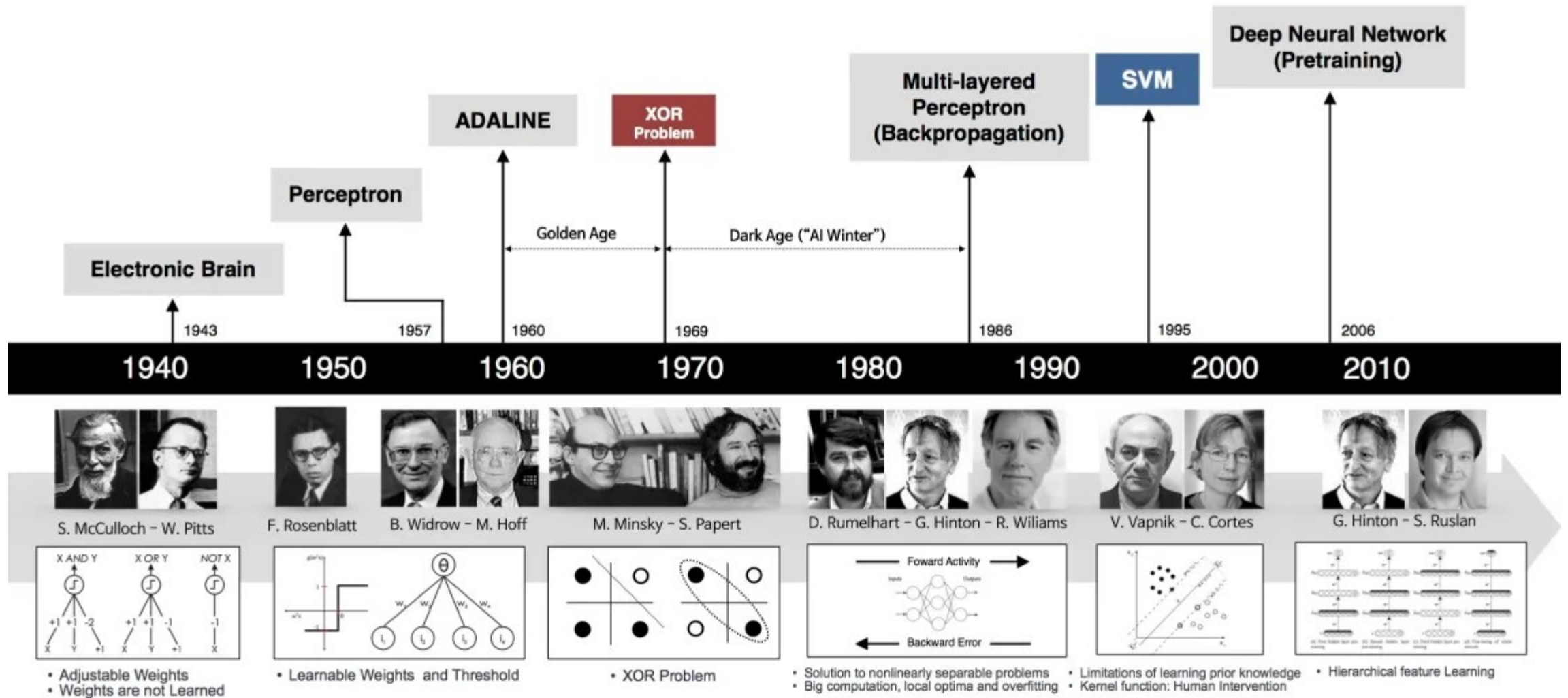


Drivers of AI - Compute

- Many of the ideas of modern AI were developed as far back as 1947, neural networks for example
 - Implementing them was beyond the capability of computing capabilities
- The first requirement for AI to be successful is enough computing power to build the models and systems
 - This has been one of the main enablers of modern LLM and GAI development
- Current hot issue in LLMs is whether or not development can be sustained because of resource usage

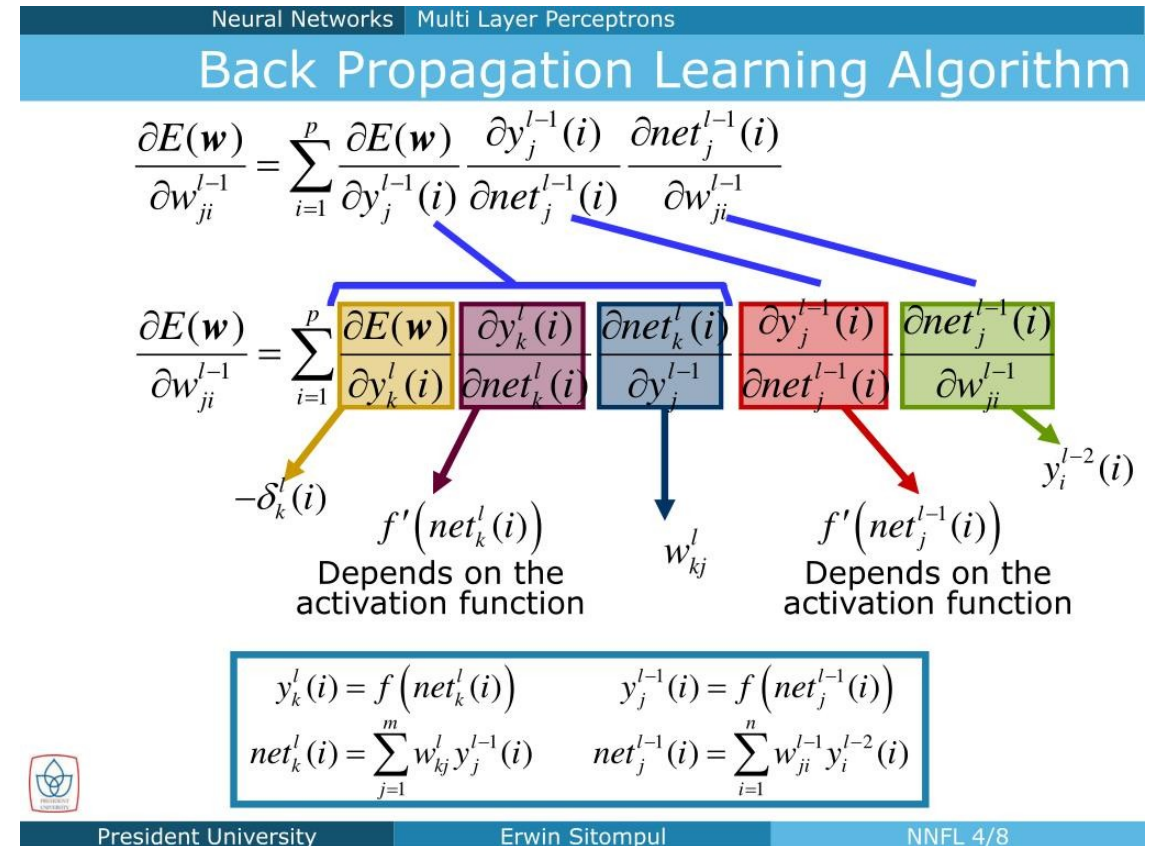


Neural Networks Timeline



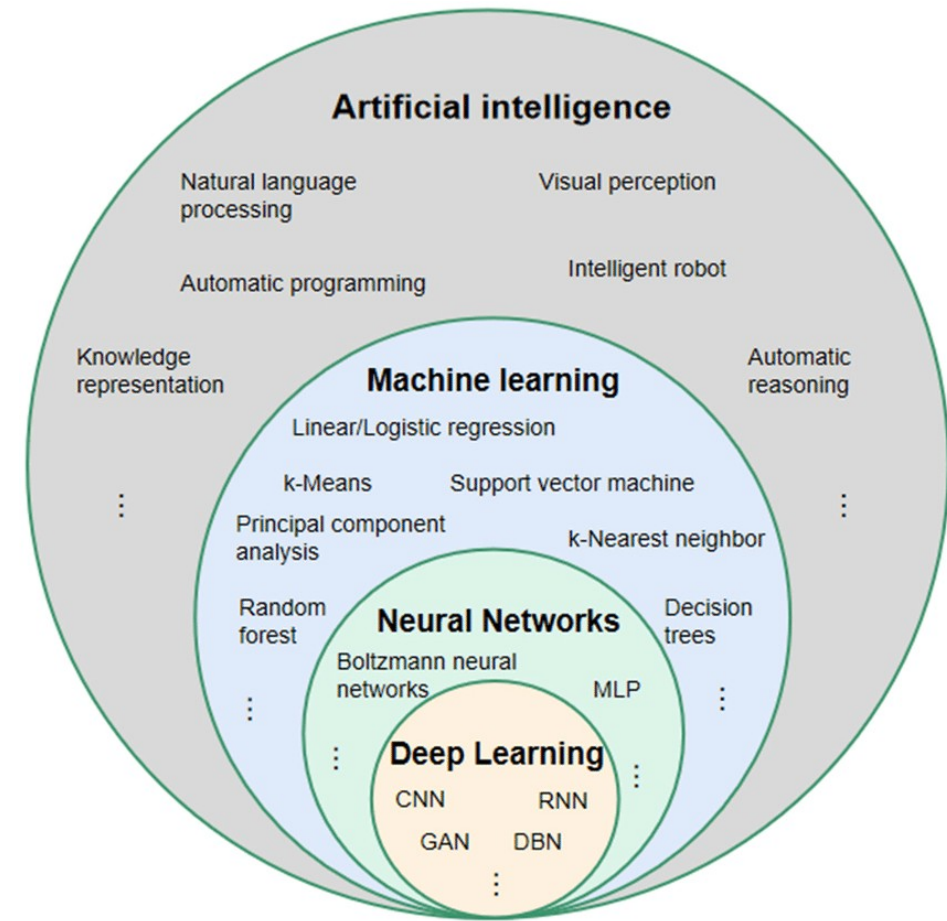
Drivers of AI - Mathematics

- The second driver is whether the mathematical tools are available
 - These are required to implement the ideas of AI into actual working models
- For example
 - Machine learning depends on the gradient descent algorithm
 - Training neural networks depends on the back propagation algorithm



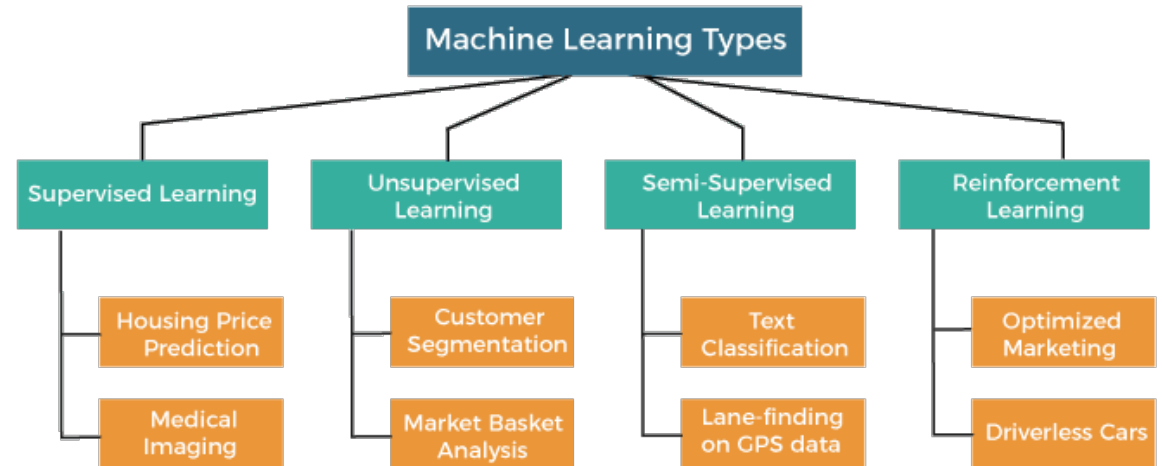
Classical Machine Learning

- Classical ML refers to statistical models created using different algorithms
 - These are drawn from other fields like statistics and data analysis
 - ML introduced techniques to create these models using “training” algorithms that worked on large data sets
 - Eg. The use of gradient descent to automatically converge to an answer
 - Eg. Measures to test how accurate and reliable the model was at making predictions
- Generally applied to a single problem
 - Spam detection in email
 - Tumour detection in medicine

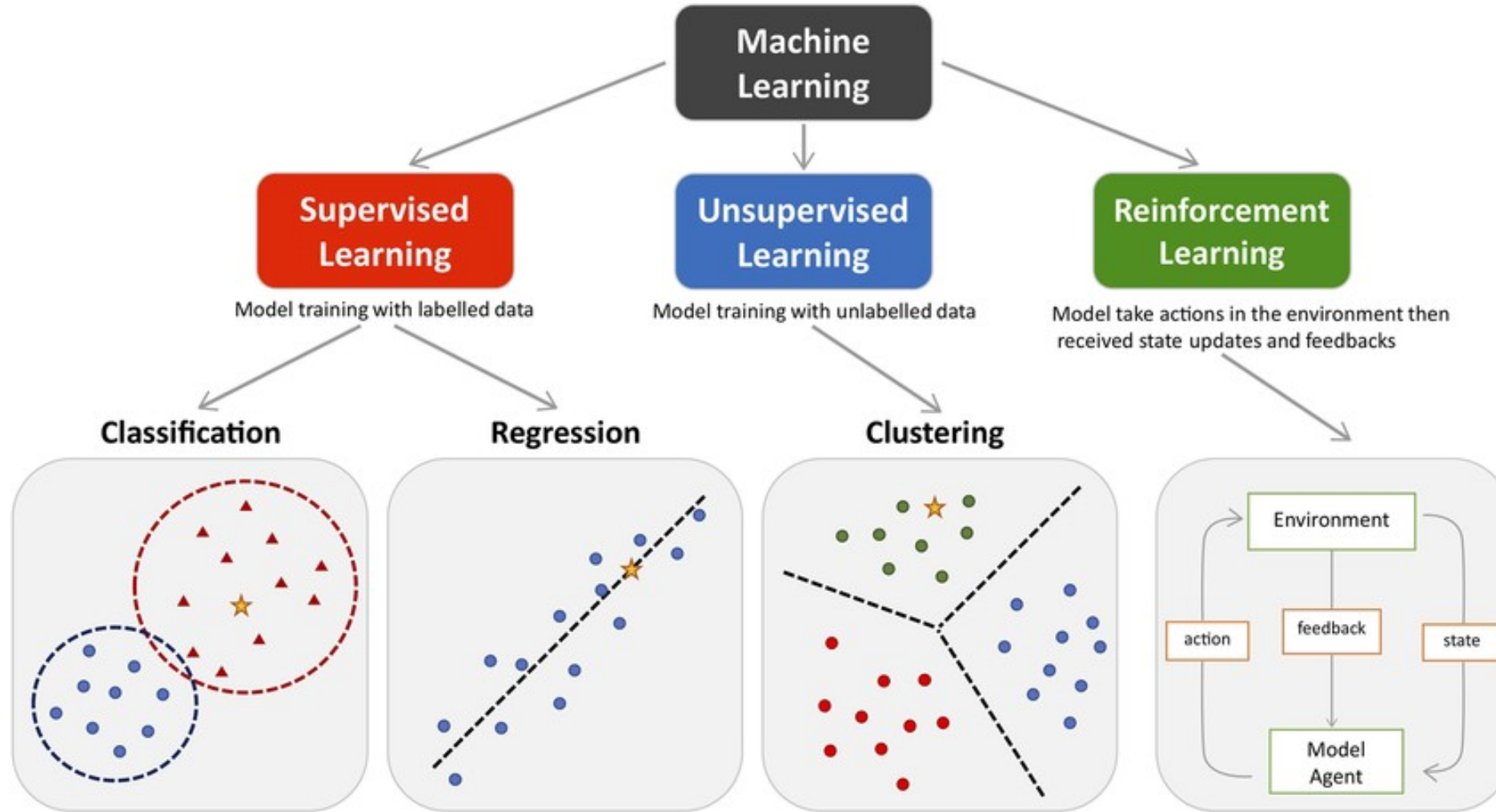


Classical Machine Learning

- Three main classical types
 - Supervised learning
 - Unsupervised learning
 - Reinforcement learning
- Semi-supervised learning
 - Later addition to the classical types that came out of deep learning
- Used for specific problems
 - Models can be developed with small data sets and limited computation
 - Much less compute intensive than neural networks

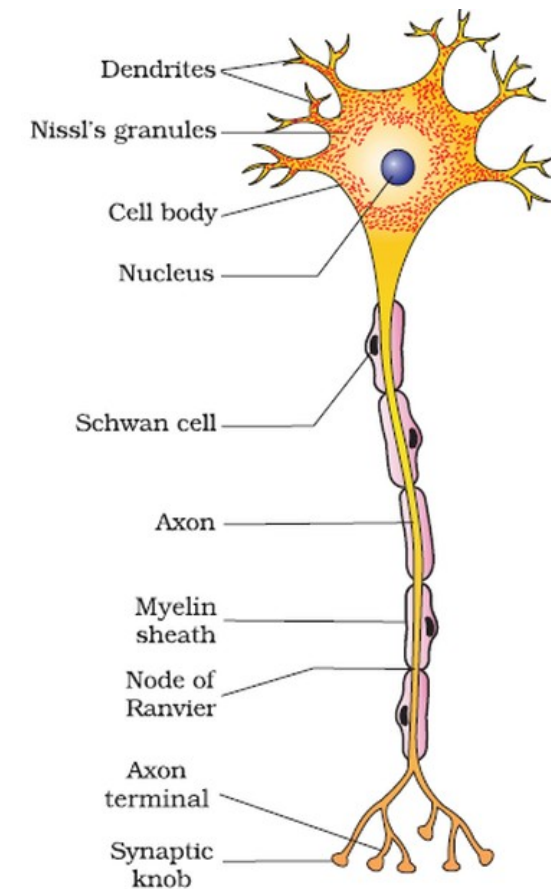


Machine Learning Typology



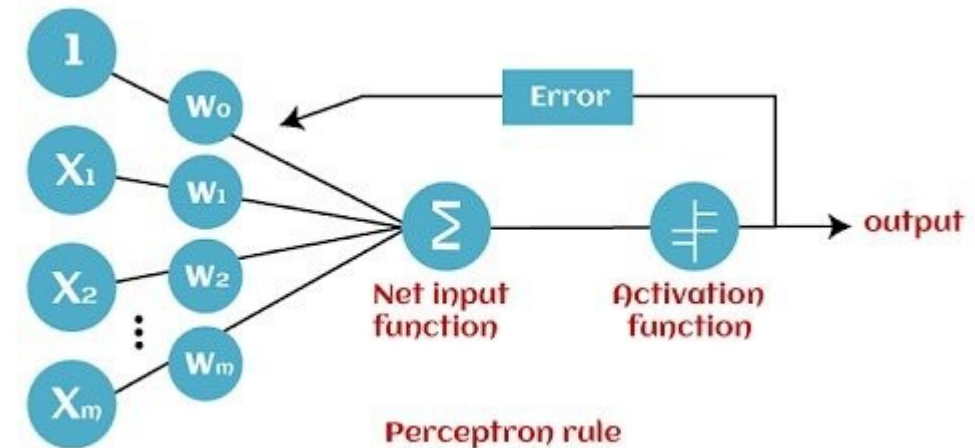
Neural Networks

- Artificial neural nets (ANN) are at the core of Deep Learning
 - Powerful, scalable and can solve complex problems like classifying billions of images
 - ANNs inspired by neurons in human brain
 - Original theoretical formulation in the 1940s
- First working models in the 1960s
 - Very limited in power and could not simulate actual neurons
 - But early work provided an architectural model for deep learning
 - No longer intended to simulate human neurons



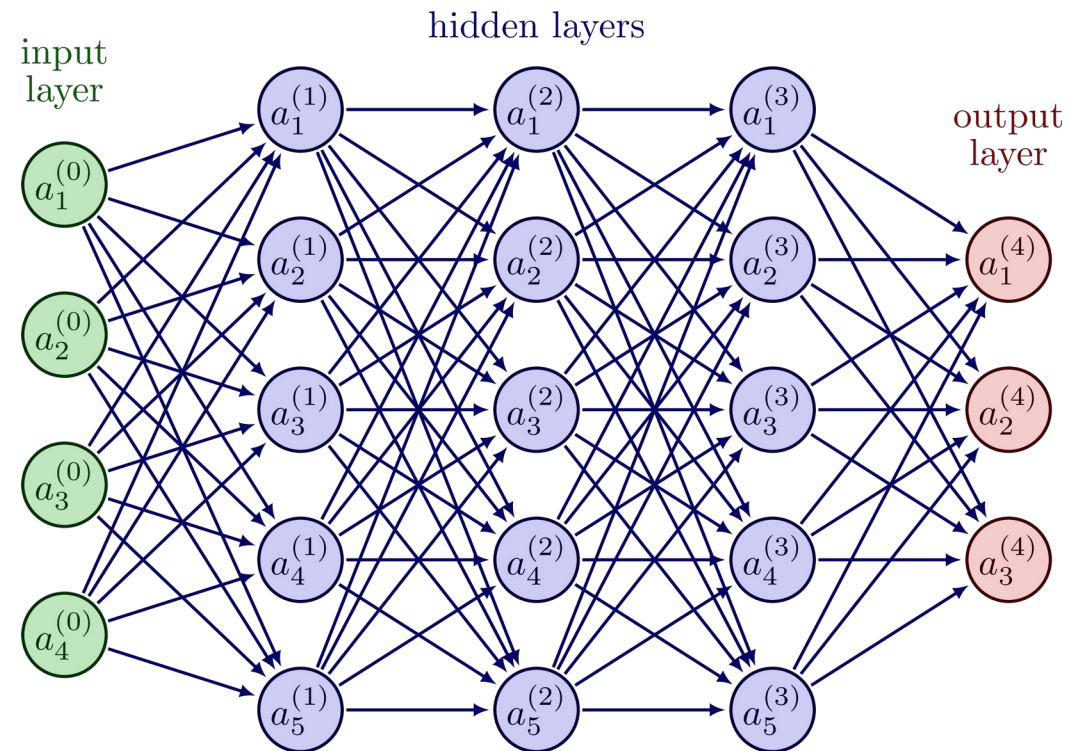
Deep Learning

- 1943: McCulloch Pitts neural model
- 1962: Frank Rosenblatt invented the perceptron
- 1969: Minsky's paper killed interest in ANNs
 - Demonstrated the ANNs can't solve a simple XOR problem
- 1970s: No work on ANNs - first AI winter
- 1980s: Some revival in ANNs
- 1986: Rumelhart introduced backpropagation training
- 1990s: Second AI winter
 - Methods like SVMs were producing better results
 - Limited by processing power
- 2010s: Huge revival in AI after some promising results



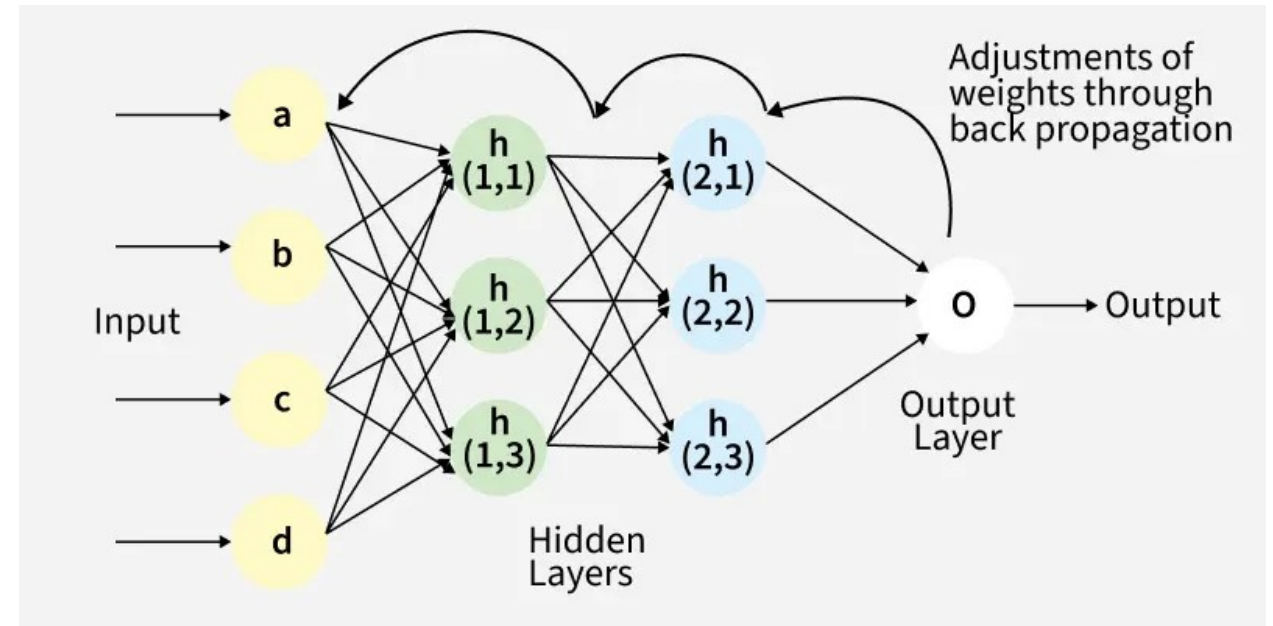
Feed Forward Nets

- Have are multiple layers
 - Each layer has many neurons
 - Each neuron is connected to neurons on previous layer
- Information flows through in one direction - no feedback loop
 - Made up of input, output and hidden layers
 - Nets with more than one hidden layer are called deep neural nets
- Very difficult to do any form of deep learning on a feed forward net
 - No error correction feedback possible



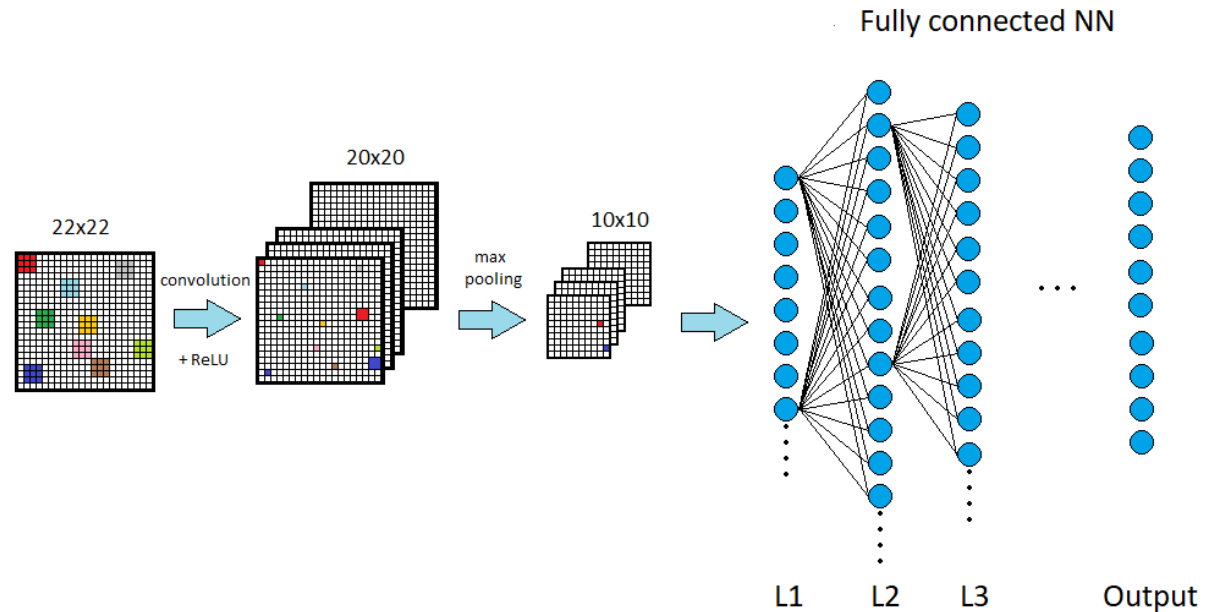
Deep Learning with Back Propagation

- Back propagation is the basis for training neural nets
 - A result is computed
 - The output is compared to the right answer
 - The last layer's inputs are "tweaked" to get it closer to the right answer
 - Then the previous layer is "tweaked" to get it closer to giving the last layer its tweaked inputs
 - This continues from back to front until all the layers have been tweaked.
- This is a very computationally expensive process and very time consuming



Convolutional NN

- CNNs were developed for image processing
- Imagine a small patch being slid across the input image
 - This sliding is called convolving
 - Similar to a flashlight moving from the top left end progressively scanning the entire image
 - This patch is called the filter/kernel and the area under the filter is the receptive field
 - The idea is to detect local features in a smaller section of the input space, section by section, to eventually cover the entire image
- These led to GAI image processing



Problems with Text

- Text processing is more difficult than image processing
- To understand text, the system has to remember what it read previously
 - It has to remember the context of the discourse
- Needed for
 - Disambiguation of word meanings (was the screwdriver a drink or a tool?)
 - Anaphora resolution (who does “he” refer to?)
 - Basic understanding of what the text is about

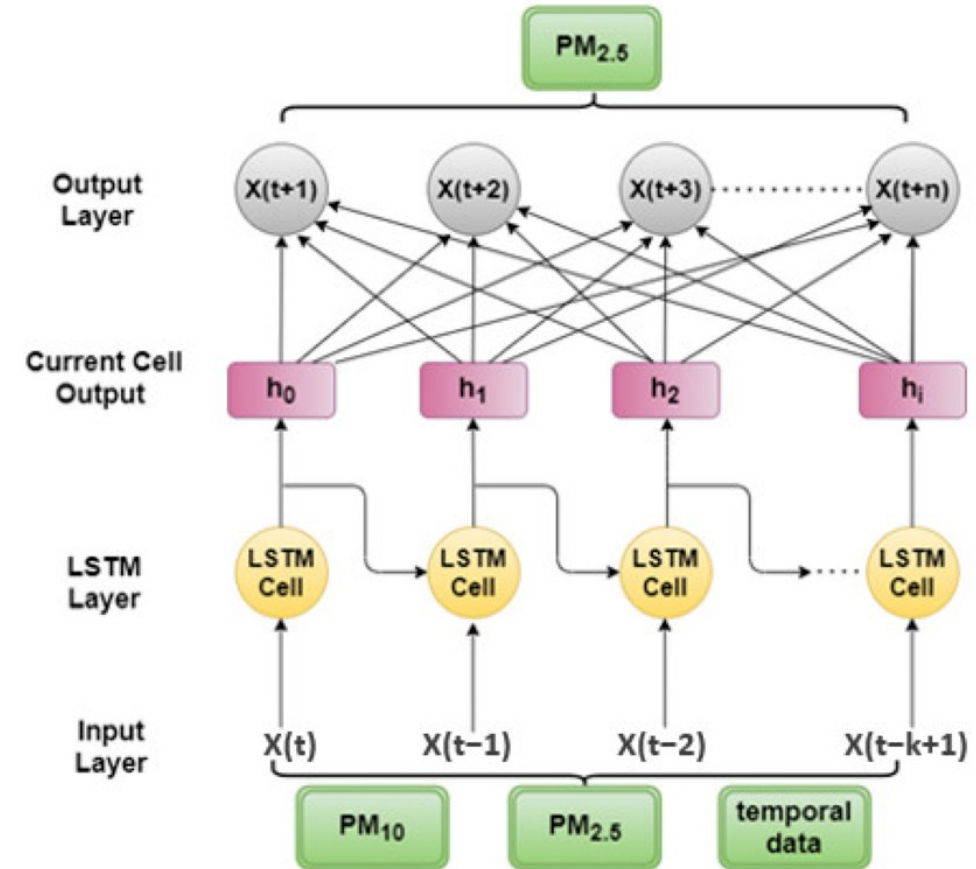
Bob had always hated Dave. He always had.

One day, he was drinking at his favourite bar when he walked in. Furious that he would show up at his favourite bar, he threw a screwdriver in his face.

Dave wasn't hurt so he declined to press charges.

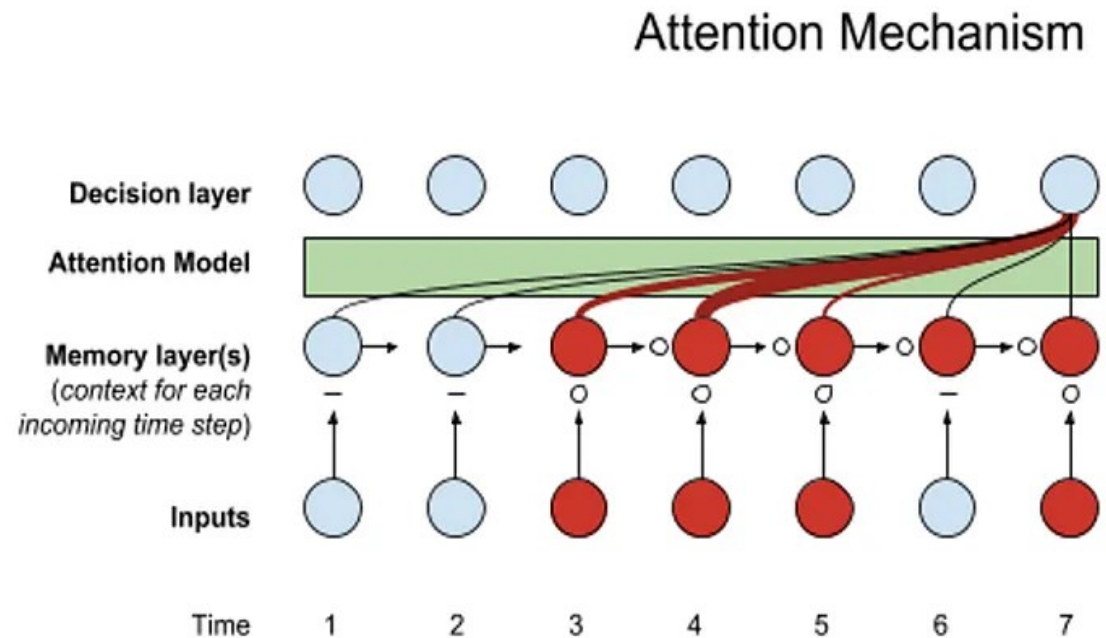
NN with Memory

- Early attempts at text processing used some form of memory to remember context
 - Recurrent NNs used a feedback loop to feed previous information back into the net
 - LSTM tried to refine this process by adding a memory module that would “forget” older information
- This led to techniques for allowing NNs to “pay attention” to what was important in context



Attention

- When you read, not every word has the same importance
 - Focus is on certain key words that carry meaning
 - “Although he was tired, John finished the report before the deadline.”
 - When trying to understand who finished the report, more attention is paid to “John” and “finished” and less to filler words like “although” or “the”
 - Attention mechanisms help the model focus more on the important words when making a decision or prediction
 - “Not all words matter equally, so pay more attention to these ones that do matter”
 - It requires analyzing a sentence and its context at the same time looking for patterns of occurrence
 - And then developing relationships in the model to reflect the patterns in the text



Transformers

- Transformers read an entire sentence, or paragraph, at once
 - They can figure out which words relate to each other, no matter where they appear
- “The bank approved the loan because the customer had excellent credit”
 - A transformer
 - Understands that “approved” is connected to “bank”
 - Connects “excellent credit” with the reason for approval
 - Doesn’t get confused if you rearrange the sentence slightly
 - Uses attention to weigh the importance of each word in relation to others

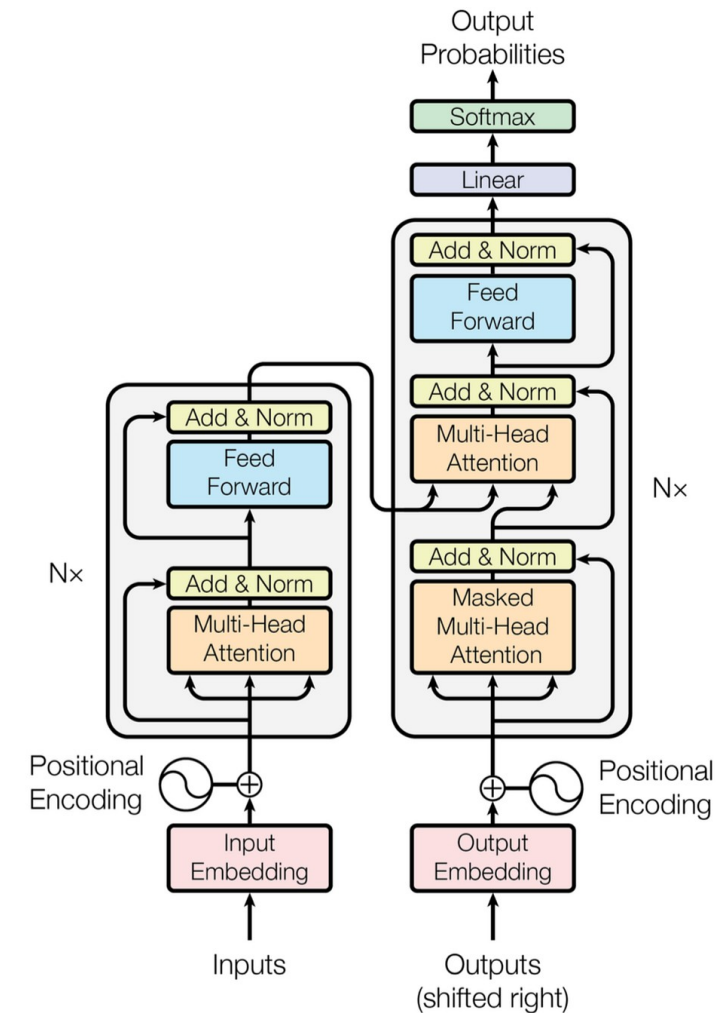


Figure 1: The Transformer - model architecture.

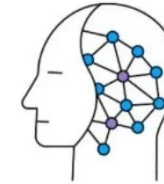
Large Language Models

- Essentially a massive transformer based statistical model language
 - Trained on massive amounts of data
 - Has billions or trillions of dimensions
 - Takes massive compute power to train, usually using specialized hardware and parallel processing
- Transformers introduced
 - Self-attention
 - Parallel processing
 - Long-range dependency modeling
 - Efficient scaling

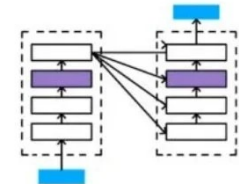
Large Language Models (LLM)



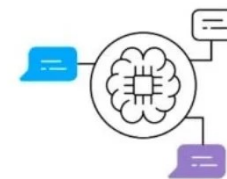
Massive Dataset



Deep Learning



Transformer Architecture



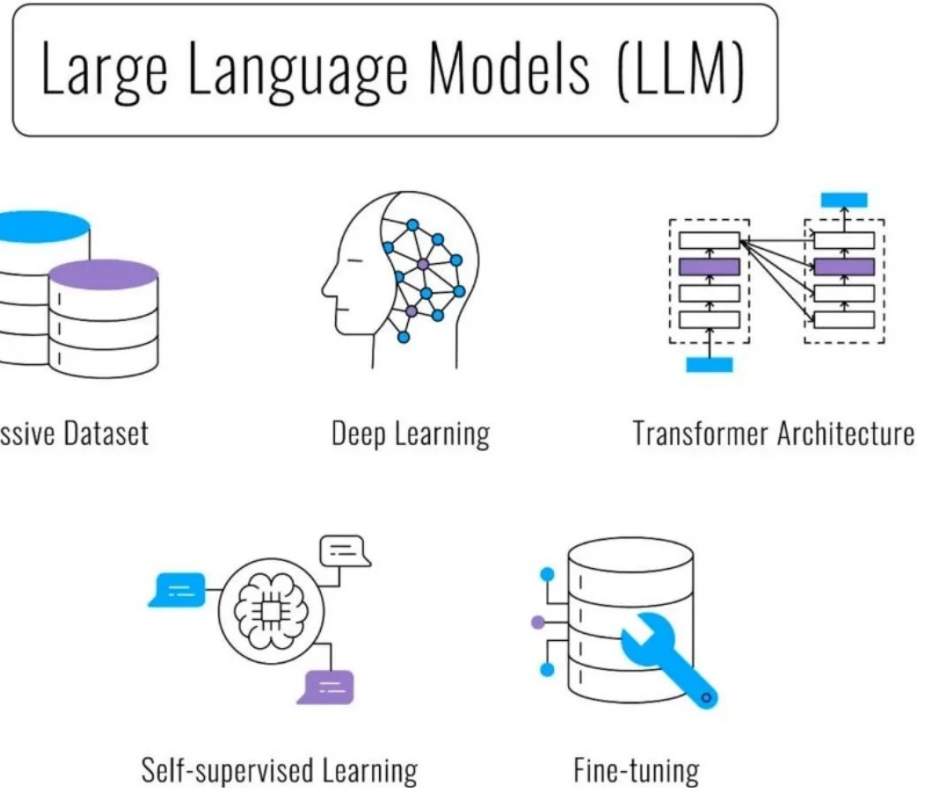
Self-supervised Learning



Fine-tuning

Large Language Models

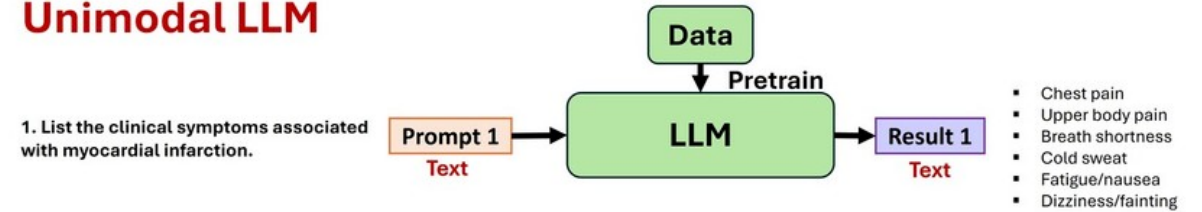
- An LLM is
 - A transformer-based neural network trained on massive text data to predict and generate language
- It combines
 - Transformer architecture
 - Large-scale pretraining
 - Next-token prediction objective
 - Fine-tuning and alignment
- We will explore these concepts in the next module



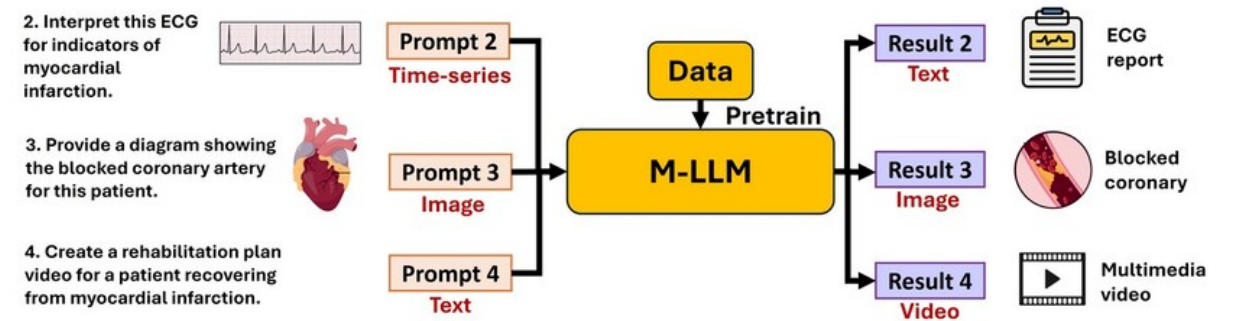
Multi-modal Training

- Multi-modal means training a model on multiple types of data
 - Text, images, audio, or video
 - Enables processing cross different types of information
 - Use cases
 - Image captioning
 - Visual question answering
 - Text-to-image generation
 - Document understanding (text + layout + images)
 - Voice assistants
 - Video understanding
 - How It Works (Conceptually)
 - Separate encoders for different modalities
 - Shared embedding space
 - Cross-modal attention mechanisms
 - Joint training objective

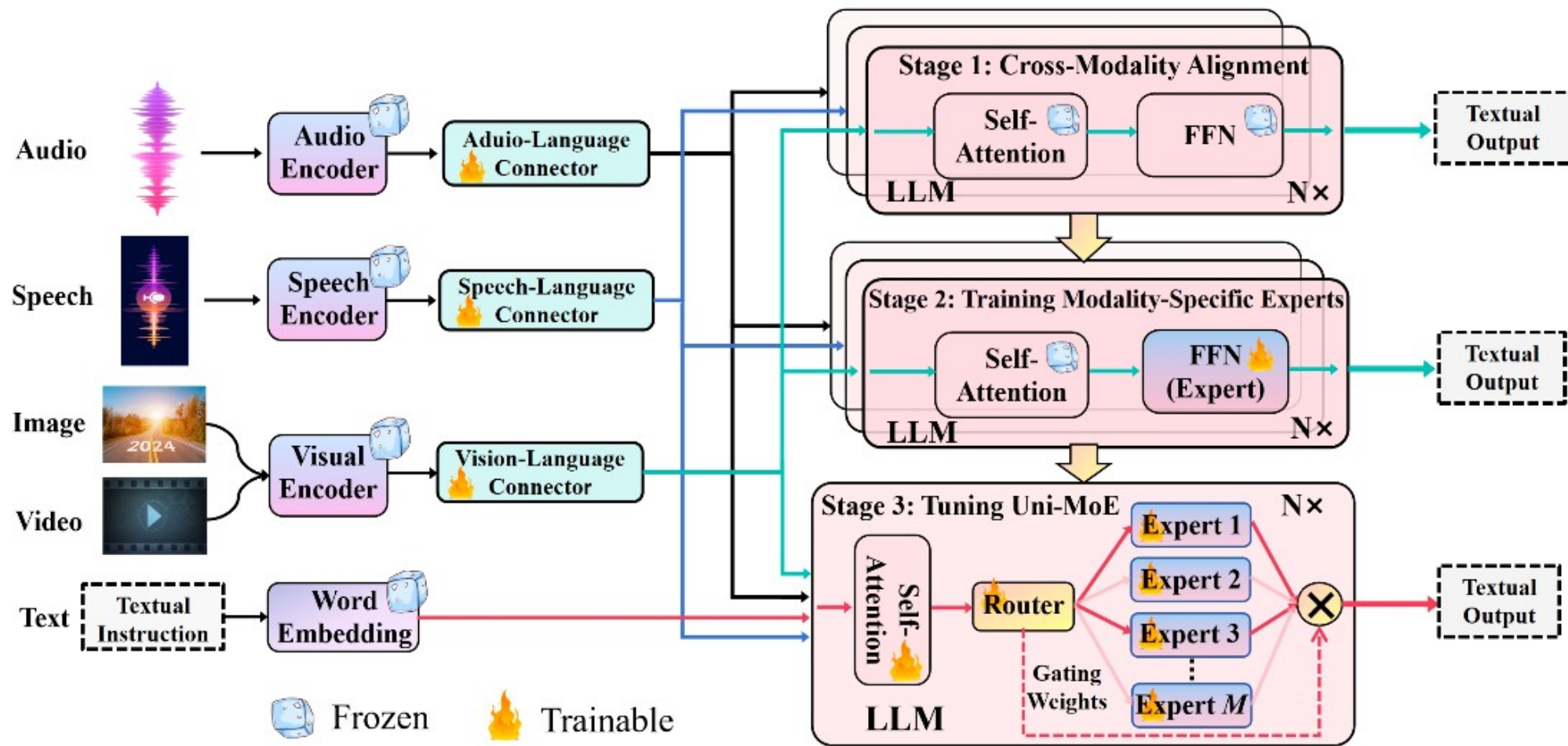
Unimodal LLM



Multimodal LLM

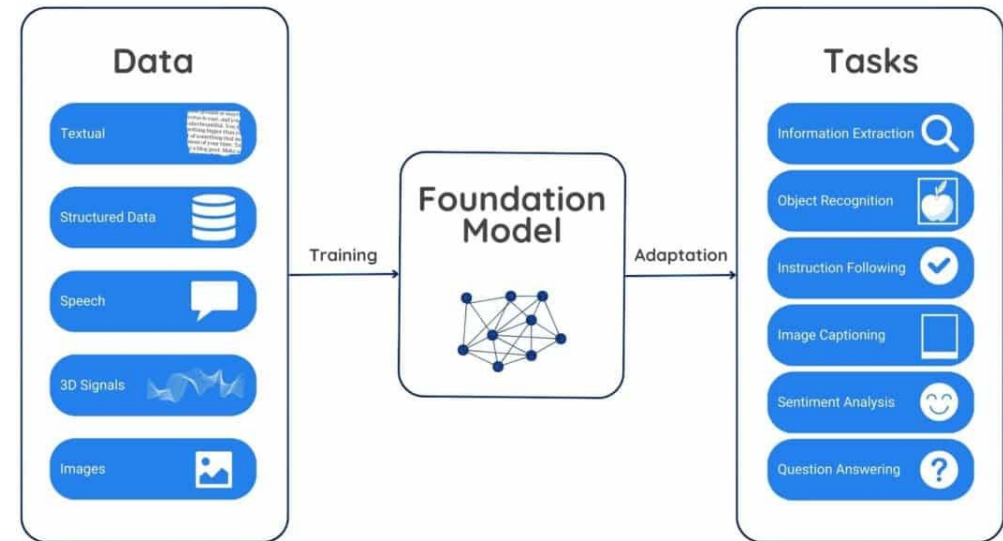


Multi-modal Training



Foundation Models

- A Foundation model is a large-scale model trained on broad multi-modal data that can be adapted to many downstream tasks
- Basic characteristics
 - Trained on massive, diverse datasets
 - General-purpose rather than task-specific
 - Can be fine-tuned to adapt the model to different tasks or context
- Serves as a base (“foundation”) for many applications
 - The term LLM is often used when people are referring to a foundation model

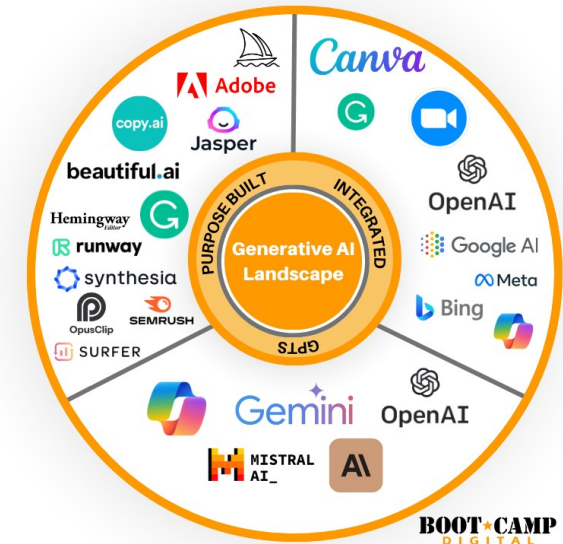


Traditional ML vs LLM vs Foundation Model

Dimension	Traditional ML	Large Language Model (LLM)	Foundation Model
Scope	Task-specific	Language-focused	General-purpose
Training Data	Labeled, narrow dataset	Massive text corpora	Massive, broad datasets (text, images, audio, etc.)
Training Objective	Optimize for a specific task	Next-token prediction	Broad pretraining objective (often self-supervised)
Model Size	Small to medium	Billions+ parameters	Billions to trillions of parameters
Architecture	Many types (trees, SVM, CNN, etc.)	Transformer-based	Usually transformer-based (or diffusion for vision)
Adaptability	Must retrain per task	Prompting + fine-tuning	Prompting, fine-tuning, adapters, RAG
Generalization	Limited outside trained task	Strong within language domain	Cross-domain generalization
Data Requirement for New Task	Large labeled dataset	Few-shot / instruction	Few-shot / instruction / multimodal
Example Use	Spam classifier	ChatGPT-style chatbot	GPT-4, Claude, Stable Diffusion
Development Model	Train → Deploy	Pretrain → Align → Deploy	Pretrain → Adapt → Build ecosystem

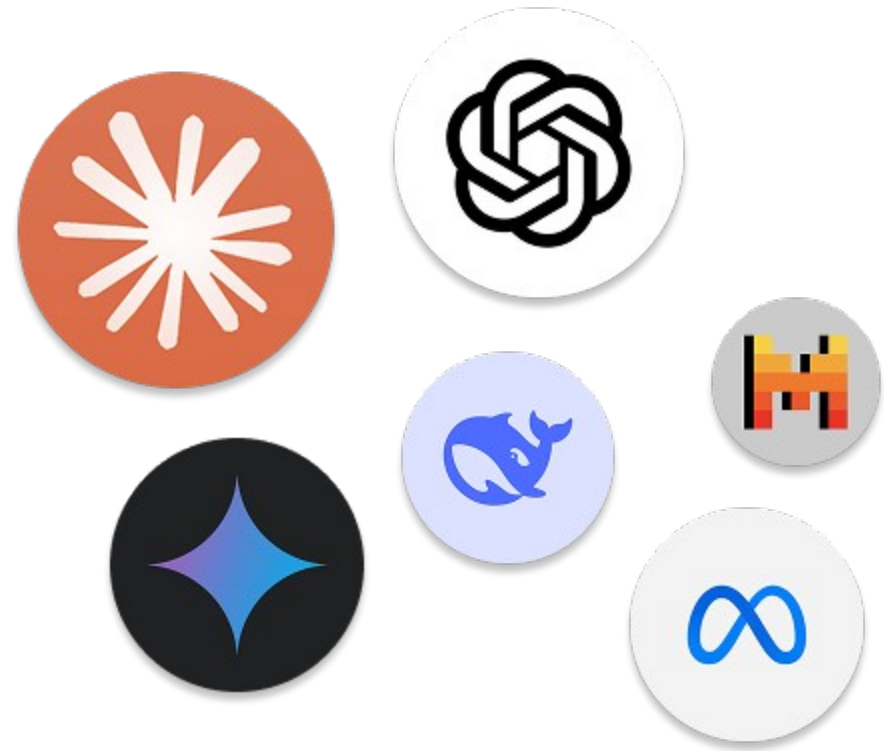
The Generative AI Ecosystem

- Generative AI today is shaped by three major forces
 - Frontier model labs (training large foundation models)
 - Cloud & infrastructure providers (compute + distribution)
 - Open-source communities (democratization + rapid innovation)
- The ecosystem is highly interconnected:
 - Labs build models
 - Cloud providers distribute them
 - Open-source drives experimentation and adoption



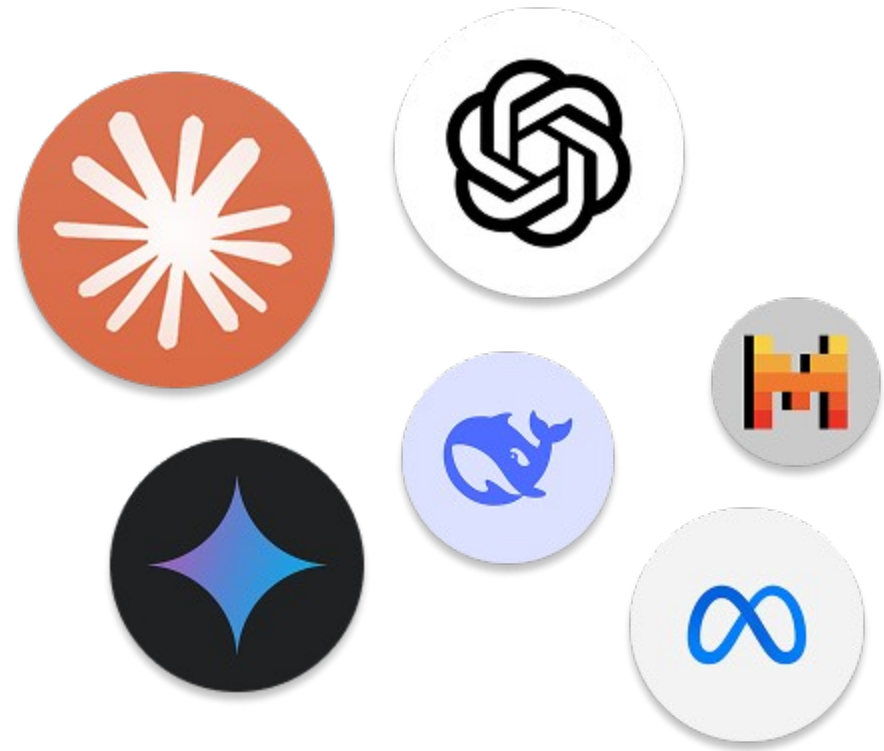
The Generative AI Ecosystem

- Frontier model labs
 - Trains state-of-the-art foundation models
 - Operates at massive computational scale
 - Invests heavily in AI research and safety
 - Works with billions to trillions of parameters
 - Pushes new capabilities
 - Reasoning, multimodality, agents, etc
- Requires enormous capital and specialized infrastructure



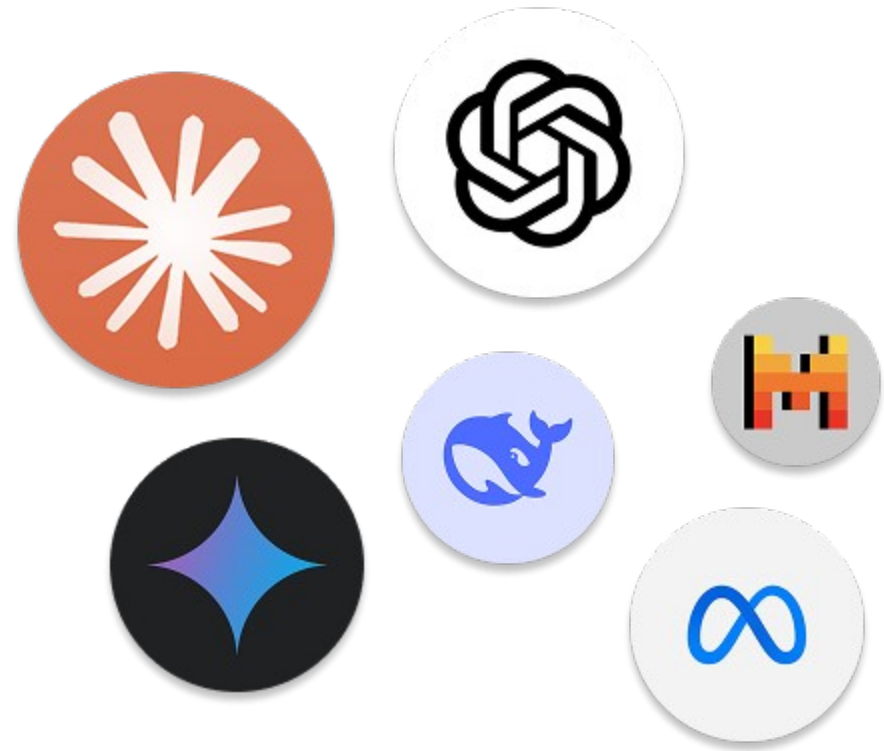
The Generative AI Ecosystem

- Frontier labs operate where
 - Costs are extremely high
 - Technical uncertainty is high
 - Capabilities are advancing rapidly
- Examples
 - OpenAI
 - Anthropic
 - Google DeepMind
 - Meta (AI research division)
 - Mistral (emerging European lab)



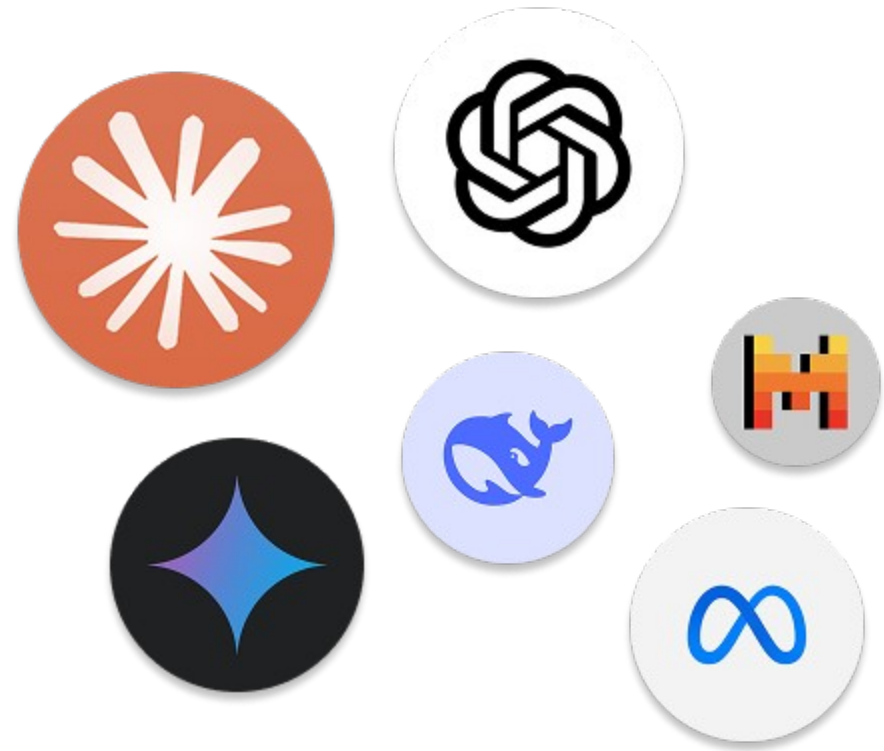
Why So Few Frontier Labs

- Training frontier models requires
 - Massive GPU clusters
 - Advanced distributed systems
 - Specialized ML researchers
 - Access to large-scale datasets
 - Billions of dollars in funding
- Only a small number of organizations can afford this.



Why Frontier Labs Matter

- Set the pace of AI capability
- Define new benchmarks
- Influence regulation and safety standards
- Shape the global AI ecosystem
- Provide the foundation models others build upon



Commercial Model Providers

- OpenAI
 - Pioneer of GPT series (GPT-3, GPT-4, GPT-4o)
 - Popularized the LLM product model (ChatGPT)
- Strong focus on
 - Multimodal models
 - Tool usage
 - Agents
 - Deep partnership with Microsoft
 - Business model: API + enterprise licensing
 - Strategic role: Set early pace in scaling LLMs commercially.



Commercial Model Providers

- Anthropic
 - Creator of Claude models
- Strong emphasis on
 - AI safety
 - Constitutional AI
 - Alignment research
 - Competes directly with OpenAI in enterprise LLM space
 - Partnerships with Amazon (AWS Investment)
 - Strategic role: Safety-focused frontier model lab



Commercial Model Providers

- Google DeepMind
 - Gemini family of models
- Long research history in
 - Transformers
 - Reinforcement learning
 - AlphaGo, AlphaFold
 - Strong multimodal capabilities
 - Deep integration with Google products
 - Strategic role: Research powerhouse + platform integration.



Commercial Model Providers

- Meta
 - LLaMA model family
 - Open-weight strategy
- Focus on
 - Open ecosystem
 - Developer adoption
 - Major driver of open model competition
 - Strategic role: Accelerating open-weight ecosystem.



Commercial Model Providers

- Microsoft
 - Infrastructure provider (Azure AI)
 - Strategic investor in OpenAI
- Embeds LLMs into
 - Office
 - GitHub Copilot
 - Enterprise stack
 - Focus on AI as enterprise productivity layer
- Strategic role: Distribution + enterprise integration.



Commercial Model Providers

- Amazon
 - Bedrock platform (multi-model access)
 - Anthropic partnership
 - AWS infrastructure dominance
- Focus on
 - Enterprise AI infrastructure
 - Model hosting and orchestration
 - Strategic role: AI as cloud service.



Amazon Bedrock

Commercial Model Providers

- Mistral
 - European frontier lab
 - High-performance open-weight models
- Strong in
 - Efficient architectures
 - MoE (Mixture of Experts)
 - Positioned as independent alternative to US giants
 - Strategic role: Efficient, competitive open-weight models.



Open-Source Ecosystem

- Hugging Face
 - Central hub for open models
 - Model hosting + sharing platform
 - Transformers library
 - Datasets + evaluation tools
 - Community-driven experimentation
 - Strategic role: GitHub of AI models.



Hugging Face

Open-Source Ecosystem

- LLaMA Family (Meta)
 - Open-weight large language models
 - Spawned thousands of derivatives
 - Enabled startups to fine-tune powerful models
 - Backbone of many open-source AI systems
 - Impact: Democratized high-quality LLMs.



Open-Source Ecosystem

- Stable Diffusion
 - Open-weight diffusion model for image generation
- Triggered explosion in
 - AI art tools
 - Custom fine-tuning
 - Local model usage
 - Major shift from proprietary-only image generation
 - Impact: Brought generative vision to the masses.



Open-Source Ecosystem

- Falcon, Mistral (Open Models)
 - Falcon (UAE-based TII)
 - Mistral models (European lab)
 - Competing in performance with frontier labs
 - Strong in efficient training and deployment
 - Impact: Expanding global AI competition.



What Open Means

- Open Weights
 - Model parameters are publicly available
 - Training data may not be fully disclosed
 - Training code may not be open
 - Often restrictive commercial licenses
 - Example: LLaMA

	Open Source	Closed Source
Tech Giants	 	 OpenAI  
Other Main Players	 MISTRAL AI_   Hugging Face	 cohere  AI21labs ADEPT Inflection ANTHROPIC

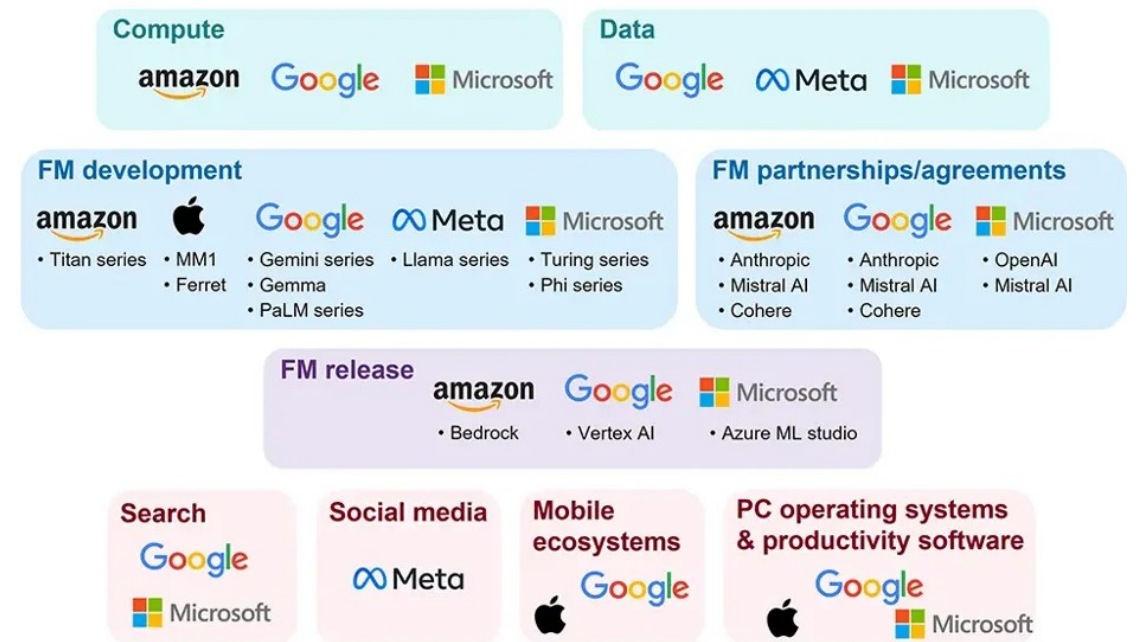
What Open Means

- Fully Open Source
 - Open weights
 - Open training code
 - Open datasets
 - Permissive licensing
 - Rare in frontier-scale models due to cost and safety concerns
- Impact
 - Controls innovation speed
 - Impacts startup ecosystem
 - Affects regulation and safety
 - Influences geopolitical AI competition

	Open Source	Closed Source
Tech Giants	 	 OpenAI  
Other Main Players	 MISTRAL AI_  stability.ai  Hugging Face	 cohere  AI21labs ADEPT Inflection ANTHROPIC

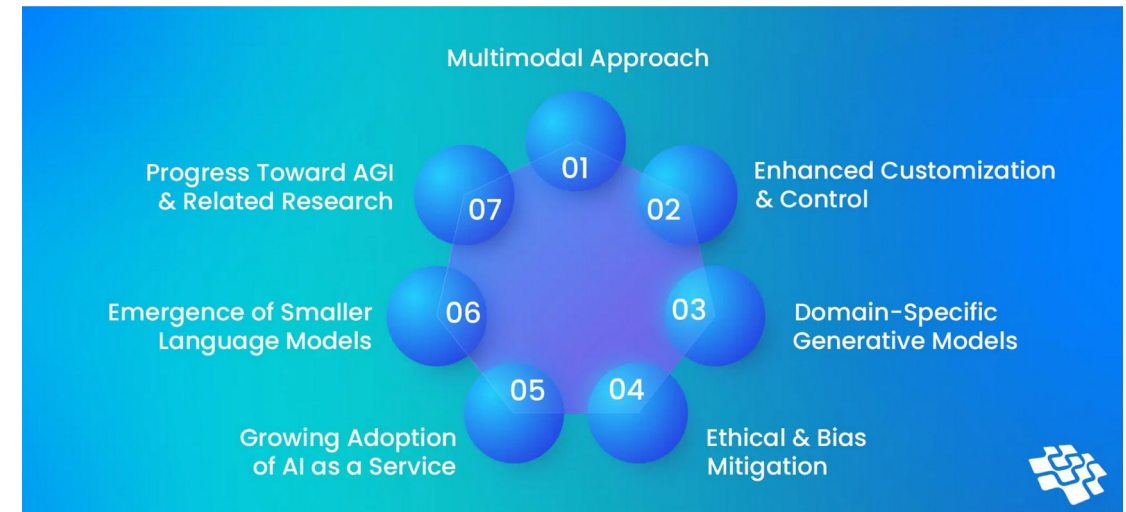
Industry Dominance

- The generative AI landscape is:
 - Oligopolistic at the frontier (few companies can train trillion-parameter models)
 - Highly competitive in open-weight space
 - Deeply dependent on cloud infrastructure
 - Increasingly shaped by safety, regulation, and geopolitics



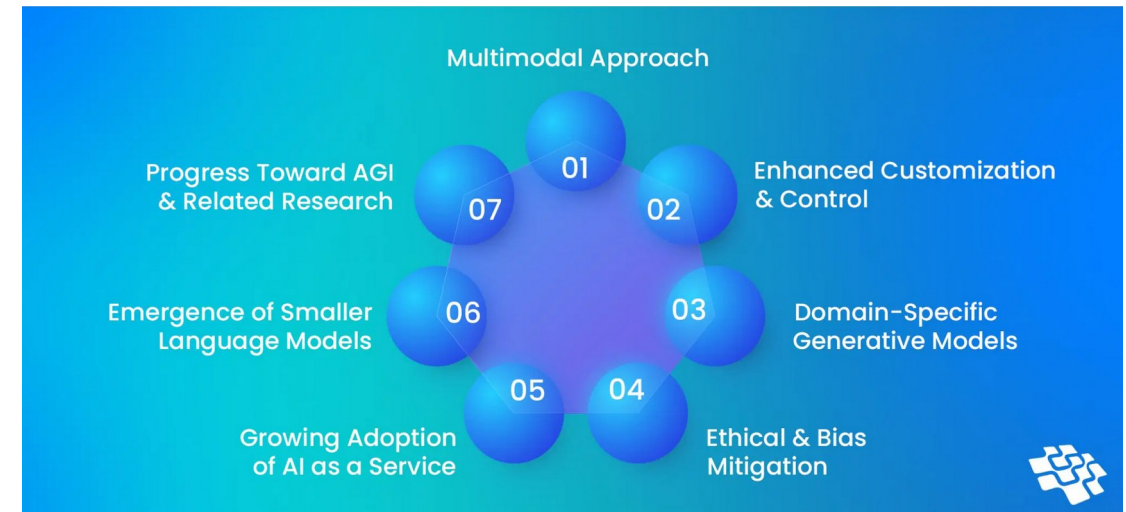
Current Trends

- Smaller and more capable models
 - Rapid improvement in
 - 3B–15B parameter models
 - Distilled frontier models
 - Quantized production-ready variants
 - Sparse / mixture-of-experts (MoE) inference optimization
 - Performance gap between
 - 70B+ models
 - Well-tuned 7B–13B models
 - is shrinking for many enterprise tasks



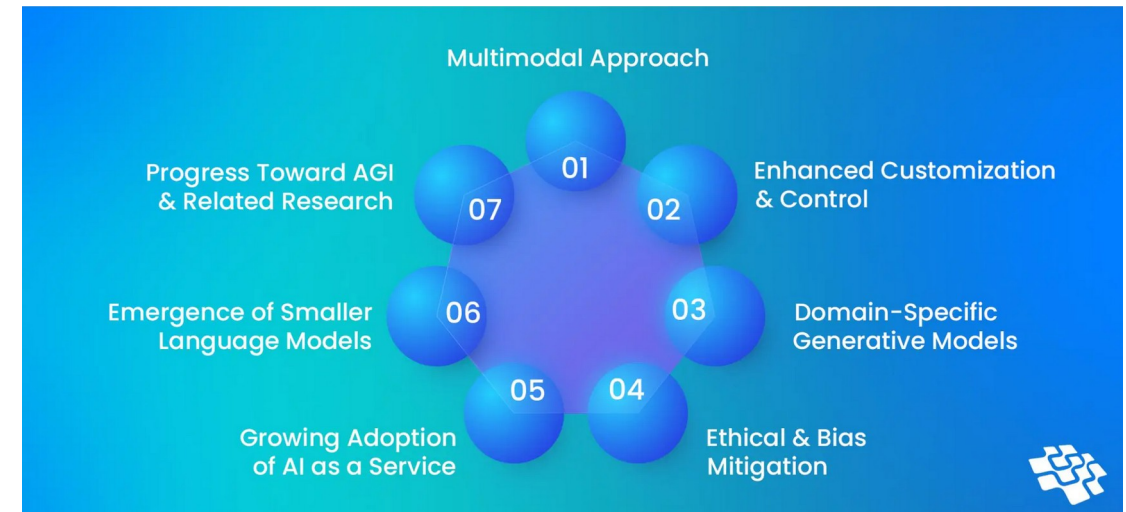
Current Trends

- Smaller and more capable models
 - Within 1–2 years, many orgs will
 - Run high-quality models on-prem or VPC
 - Fine-tune smaller models cheaply with LoRA
 - Avoid relying exclusively on frontier APIs
 - This impacts
 - Infrastructure strategy
 - Cost modeling
 - Vendor lock-in decisions



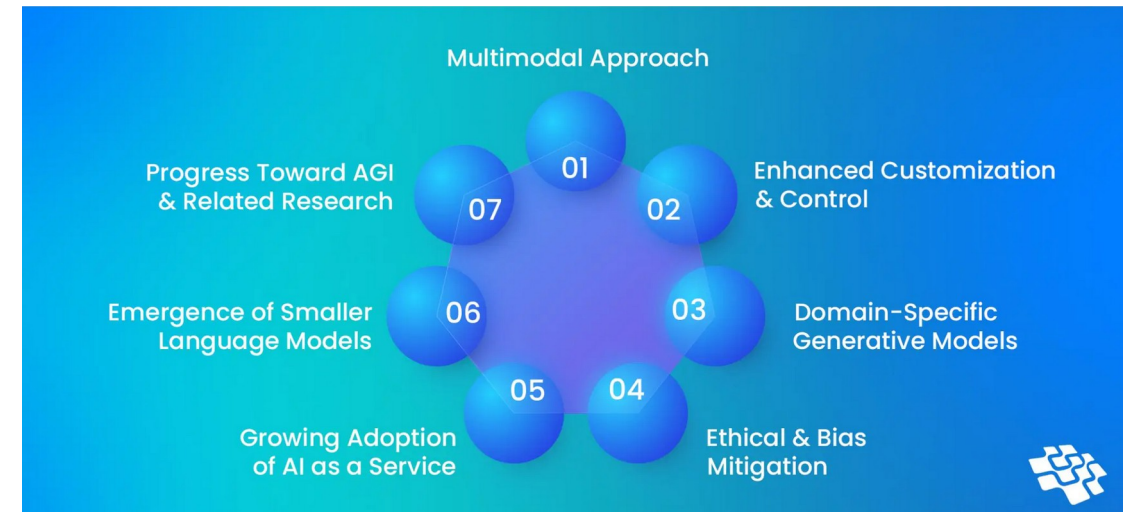
Current Trends

- Native multimodal models
 - Multimodal models are becoming:
 - Native (not stitched pipelines)
 - Better at cross-modal reasoning
 - More reliable in production
 - Within 1–2 years
 - Video understanding APIs will mature
 - Speech-to-reasoning-to-action pipelines will be standardized
 - Diagram-to-code workflows will improve significantly



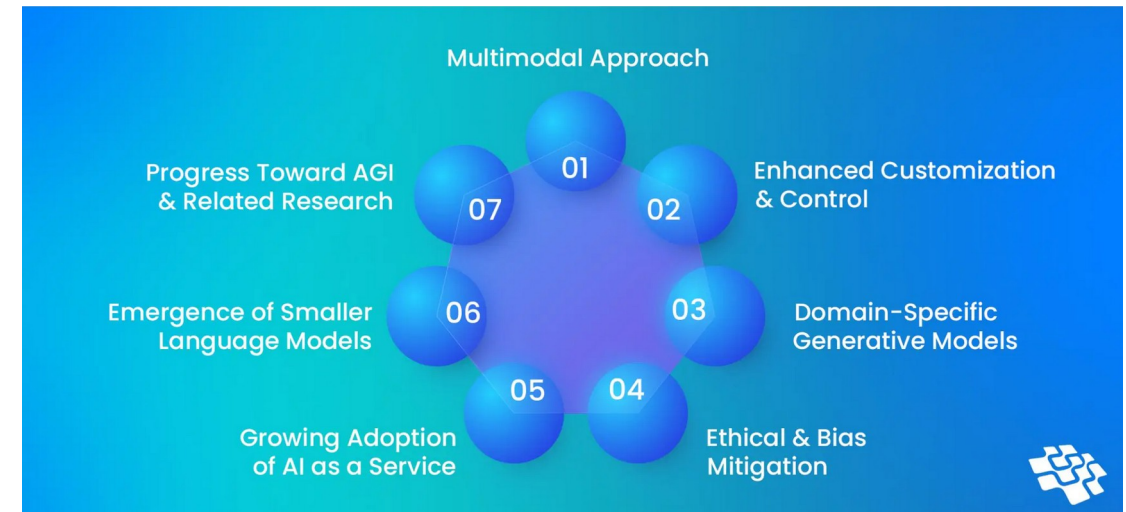
Current Trends

- Native multimodal models
 - Engineering requirements
 - Unified embedding pipelines
 - Media preprocessing layers
 - Multi-format context orchestration
 - RAG will become multimodal
 - Text documents
 - PDFs
 - Screenshots
 - Call recordings
 - Video transcripts



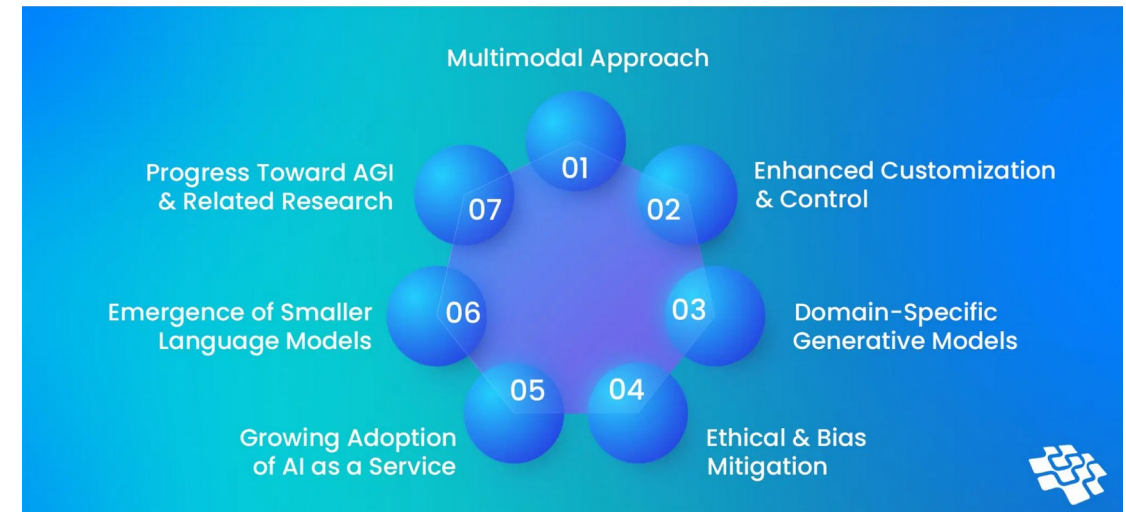
Current Trends

- Structured output – tool first
 - LLMs are shifting from “Generate text” to “Generate structured actions”
 - Trends in LLMs
 - Function calling as default
 - JSON schema enforcement
 - Tool routing layers
 - Deterministic execution wrappers
 - Within 1–2 years
 - Schema-constrained decoding to improve
 - Fewer malformed JSON outputs
 - Native validator-integrated decoding
 - LLMs become planners and not just executors



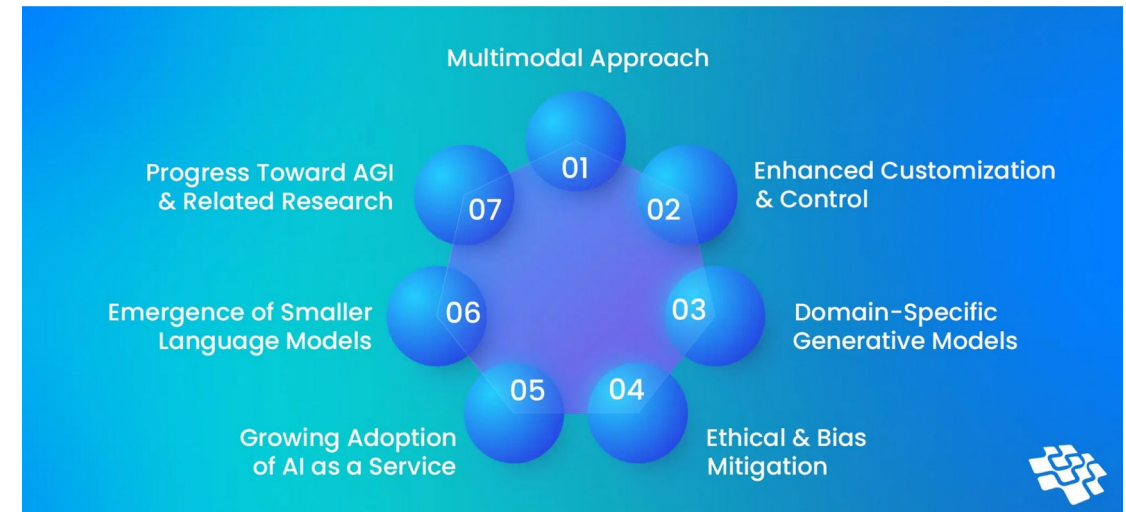
Current Trends

- Agentic systems with Guardrails
 - Agents are real but chaotic without control
 - Within 1–2 years
 - Constrained agent frameworks
 - Finite-state agent controllers
 - Graph-based execution planners
 - Hybrid symbolic + LLM systems
 - Not fully autonomous AGI-style agents
 - Controlled orchestration systems
 - What will mature
 - Multi-step reasoning pipelines
 - Long-horizon task execution with supervision
 - Retry and self-correction loops
 - Observability tooling for agents



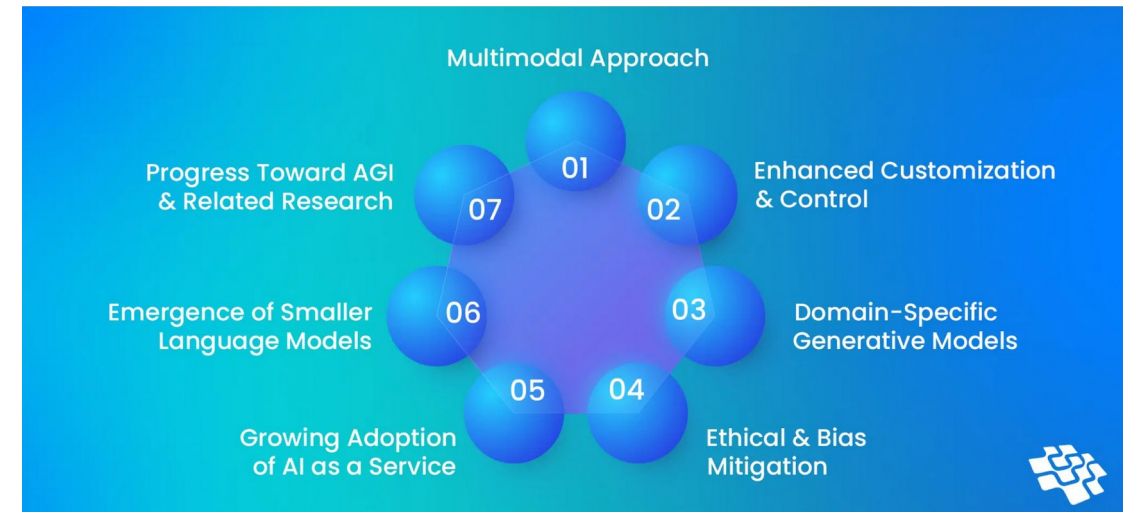
Current Trends

- Retrieval-augmented generation 2.0
 - Smarter retrieval
 - Hybrid semantic + keyword
 - Contextual reranking
 - Query decomposition
 - Adaptive retrieval depth
 - Smaller context windows
 - Instead of dumping 50k tokens, systems will
 - Retrieve less
 - Retrieve better
 - Compress retrieved content



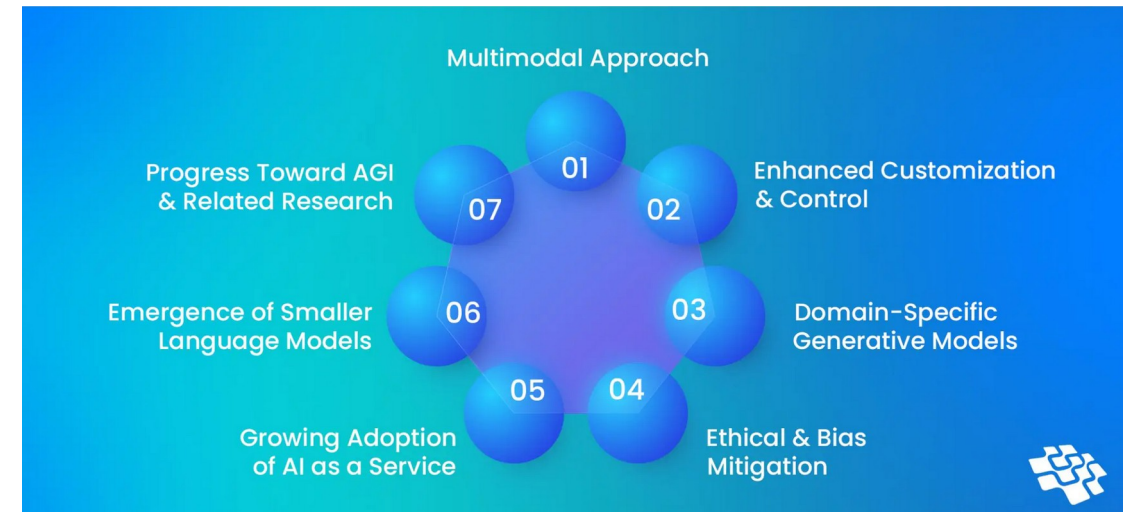
Current Trends

- Model Routing and MoE
 - One big model replaced by
 - Router models
 - Cost-aware dispatch
 - Task-specific specialization
 - For example
 - If classification then use small fine-tuned model
 - If complex reasoning then use frontier API
 - If retrieval heavy then use RAG pipeline



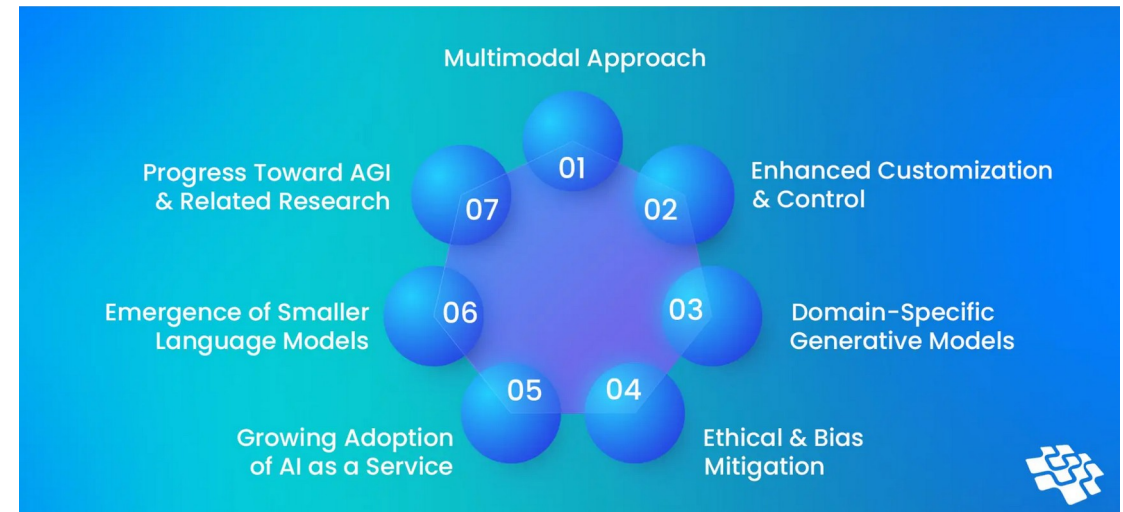
Current Trends

- AI observability and evaluation tooling
 - Right now evaluation is primitive
 - Within 1–2 years
 - Standardized LLM evaluation frameworks
 - Drift detection
 - Prompt regression testing
 - Hallucination scoring tools
 - Behavioral fingerprinting
 - This will probably be mandatory in enterprise applications



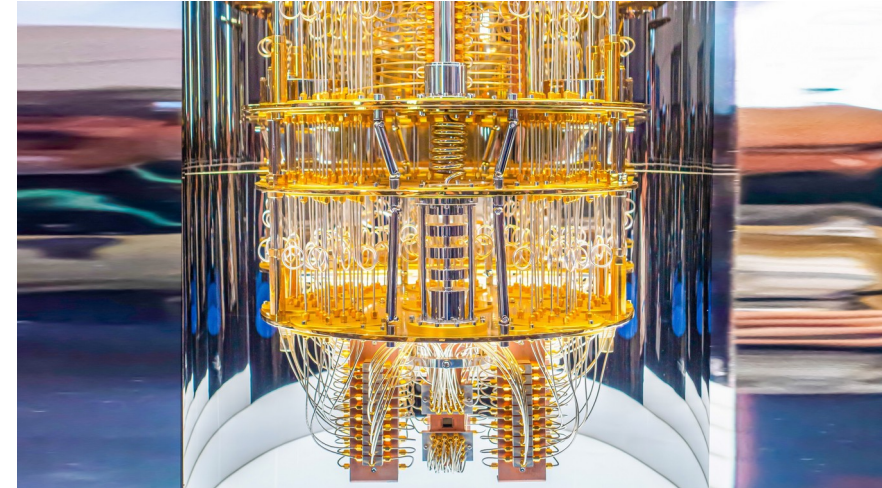
Current Trends

- AI observability and evaluation tooling
 - This will require
 - LLM test suites
 - Prompt versioning
 - Output diff tooling
 - Structured evaluation datasets
 - AI will be treated like production software, not magic



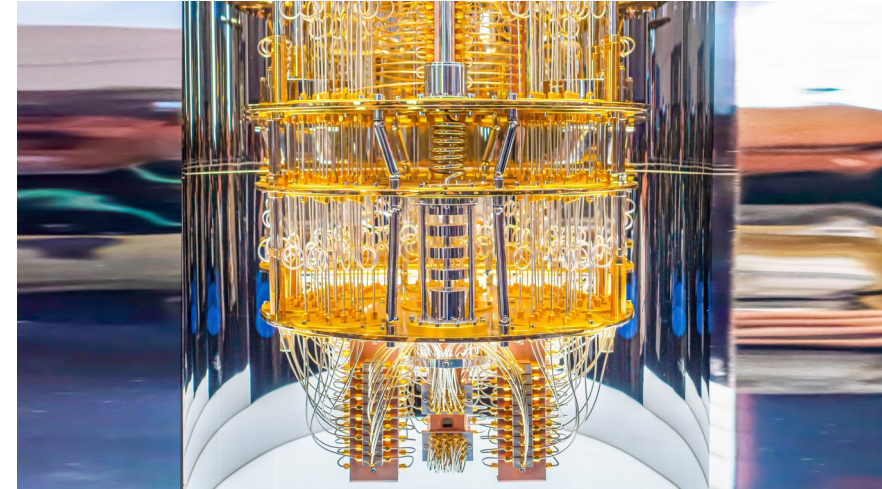
Long Term Trends

- Long term trends (10-20 years)
 - Quantum computing and ML
 - Neuromorphic hardware
 - hardware that mimics brain-like structures
 - Bio-computing, DNA-based computing
 - Most likely use: data storage
 - Silicon-biological hybrid systems
 - Photonic optical computing
 - Perform matrix multiplication using light interference
 - Near-zero energy cost per operation.
 - Extremely high parallelism



Long Term Trends

- Long term trends (10-20 years)
 - Collective AI systems
 - Consist of specialized AI swarms
 - Operate as distributed reasoning networks
 - Use consensus-based decision-making



Questions?

